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游戏设计 & 设计思维

线下研讨课-6

Tutorial-6

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目录

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- 声音
- 网络

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1 声　　音

--自强不息，止于至善--

1.1 背景

--自强不息，止于至善--

The Nature of Sound

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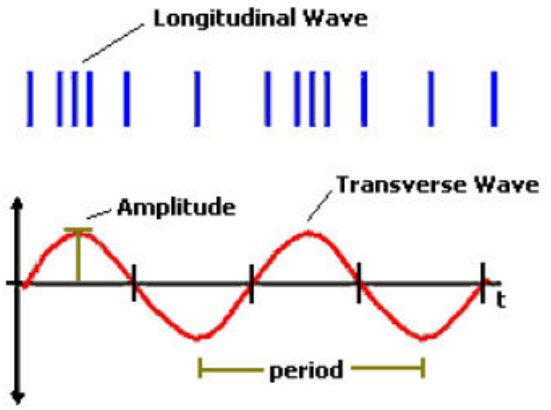
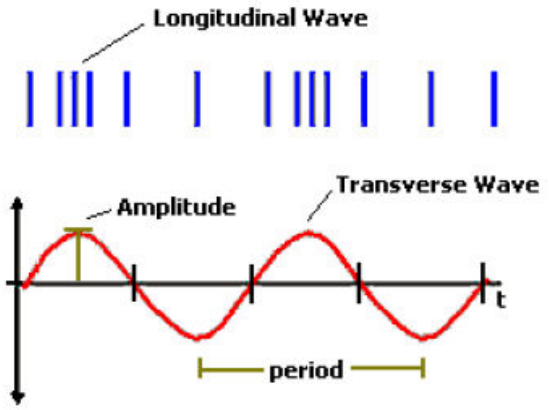
- What is Sound?:
  - Vibratory energy caused by movement of physical objects
  - As mechanical wave energy, it requires a medium such as air or water in which to propagate
- Properties:
  - Frequency: Rate of vibration
    - Experience as pitch (high or low)
  - Amplitude: Intensity of vibration.
    - Experience as loudness
    - Measured in decibels (dB) (too loud/too long hearing loss)

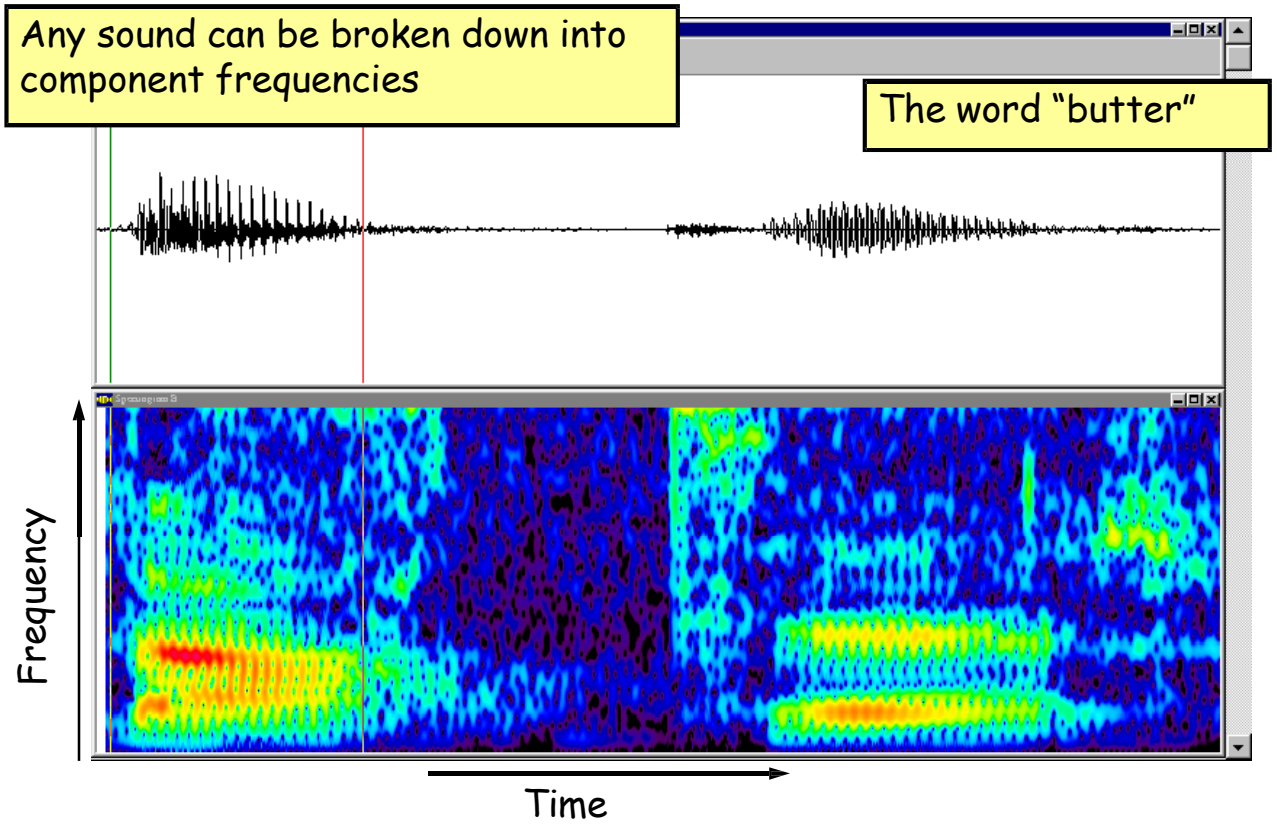


The Nature of Sound

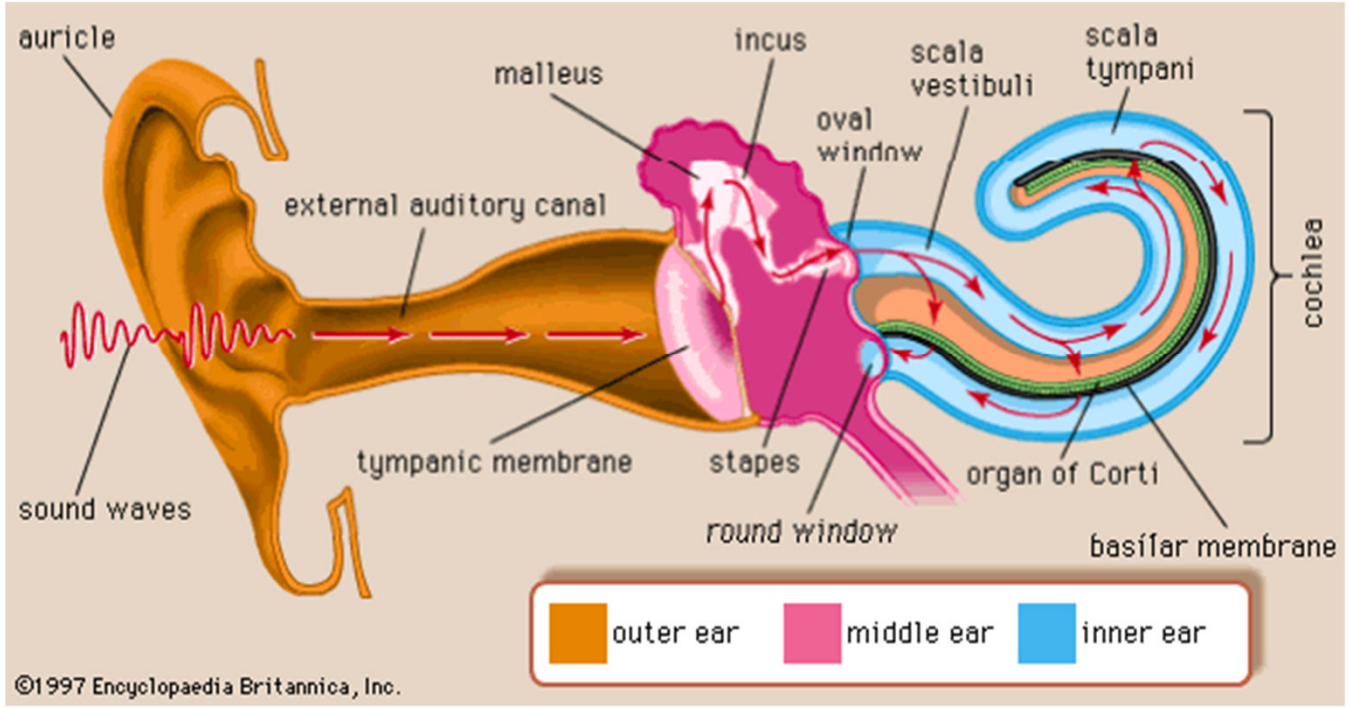
- Frequency:
  - Number of sound pulses that travel past a fixed point in a second
  - Measured in Hertz (Hz), that is, cycles per second
  - Humans: 20-20,000 Hz; Bats: 100,000 Hz
- Speed of Sound:
  - How fast sound pulse travels
  - Sound travels at same speed in a
  - given medium, irrespective of pitch
  - – 344 meters/sec in air
- Pitch:
  - As with color in vision, this is perception
  - A high frequency sound is heard as a high pitch.



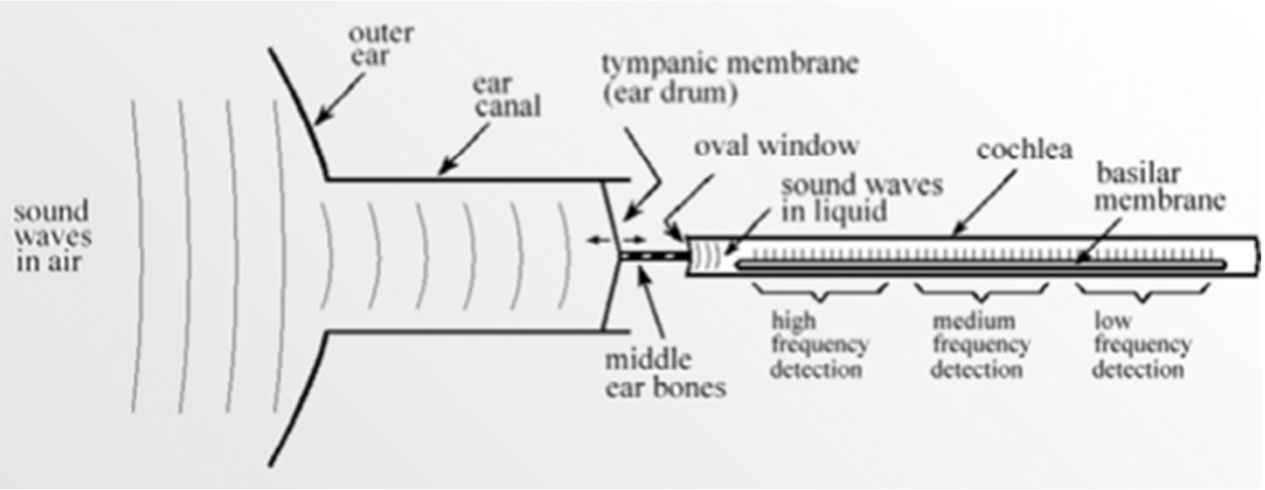
The Nature of Sound



Human Hearing



Human Hearing



- Outer ear collects sound waves from the environment
- Ear drum → middle ear bones → fluid-filled cochlea
- 12,000 nerve cells form the cochlear nerve
- Each nerve cell only responds to narrow range of audio frequencies
- The ear is a frequency spectrum analyzer

Volume

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- Perception of volume logarithmic in amplitude
- A decibel is proportional to the log of the ratio of two intensities: $10 \cdot \log10(I1/I2)$
- A 3 decibel (dB) increase is roughly equivalent to amplitude doubling
- As absolute measure, it's in comparison to threshold of audibility
  - 0 dB: inaudible
  - 60 dB: normal speech
  - 80 dB: a shout, city street traffic noise
  - 100 dB: motocycle/snowmobile
  - 120-130 dB: front row at a heavy metal rock concert
  - (AC/DC, Led Zepplin, Deep Purple)
  - 140 dB: gun blast

Pitch

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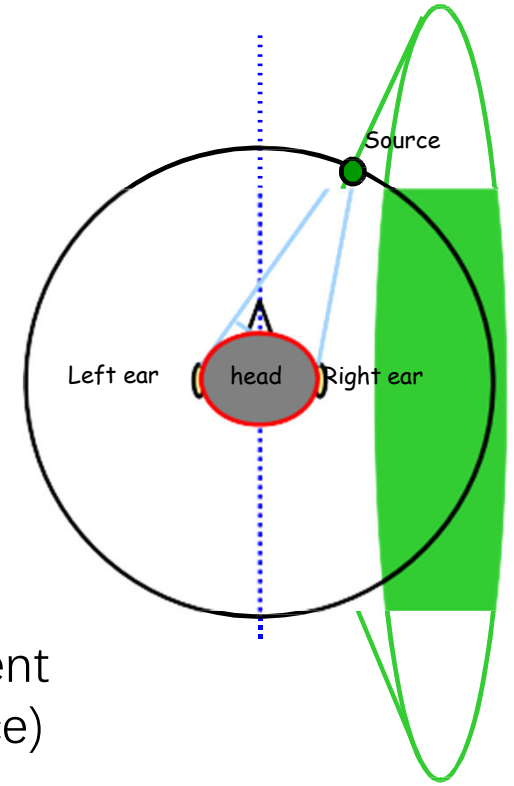
- Perception of pitch is logarithmic in frequency
- Higher frequencies are perceived as higher pitches
- Measured in hertz Hz (cycles per second) and kilohertz kHz.
- Humans audible range: roughly 5 to 20,000 Hz
- Examples:
  - 8 Hz: Lowest frequency produced by whales
  - 32 Hz: Lowest note on piano
  - 250 Hz: Typical speaking voice
  - 440 Hz: Baby's cry
  - 4096 Hz: Highest note on a piano
  - 200,000 Hz: Highest frequency produced by dolphins

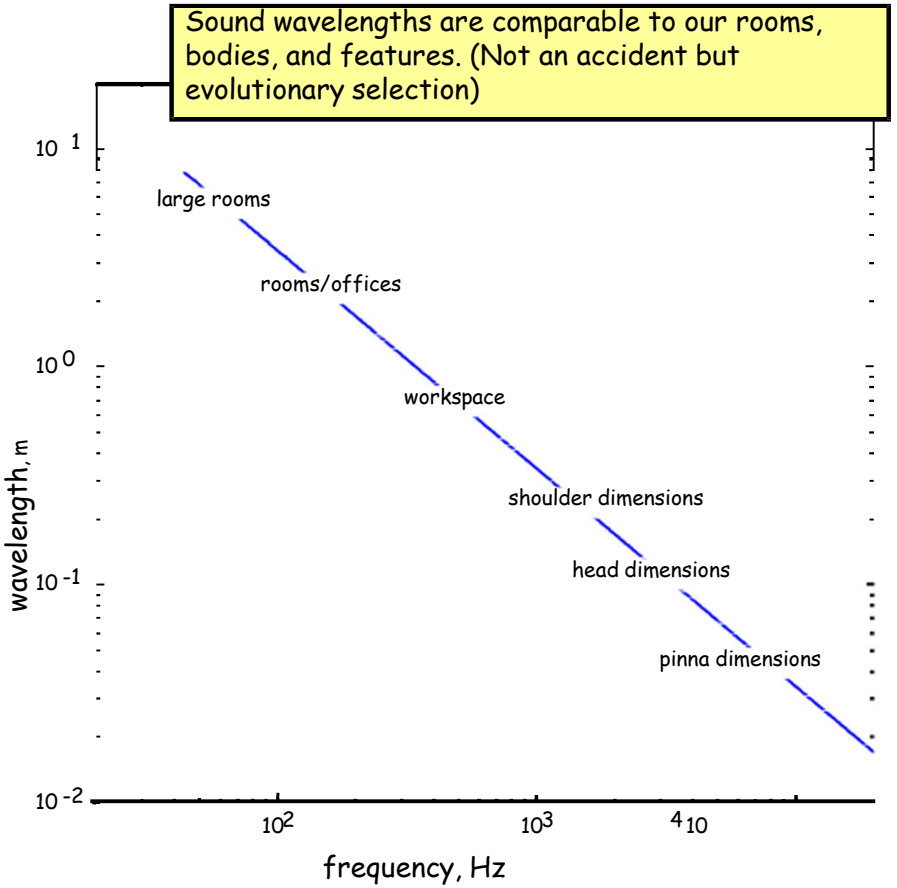
How do we perceive sound location?

- Initial idea:
  - Measure attributes of received sound at the two ears
- Compare sound received at two ears:
  - Interaural Level Differences (ILD)
  - Interaural Time Differences (ITD)
- Surfaces of constant Time Delay:
  - $| x - xL| - |x - xR| = c \cdot \delta \cdot t$
  - Delays same for points on cone-of-confusion
  - Level differences are small
- Other mechanisms necessary:
  - Scattering of sound off our bodies and off the environment
  - Purposive motion (e.g. moving left-right to localize source)



Sound and Human Spaces

- Sound Wavelengths:
  - Comparable to human dimensions and of the spaces we live in
  - Wavelength λ is inversely proportional to frequency f
- Cases: a = object size
  - λ >> a: the wave is not affected by object
  - λ ~ a: scattered wave is complex and diffraction effects are important
  - λ << a: wave behaves like a ray



Modeling Sound Scattering

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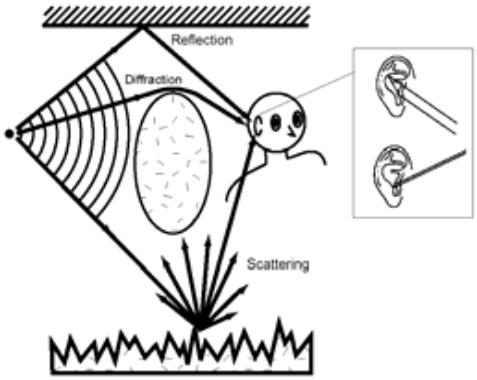
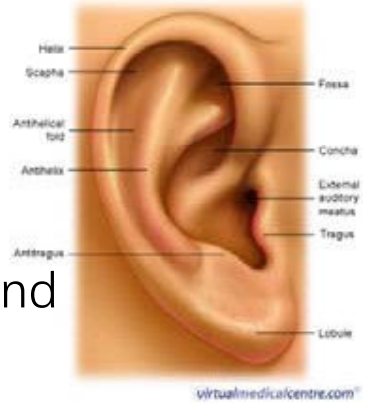
- Interactions change received sound waves.
- Scattering of body and ears:
  - Bodies ~ 50 cm
  - Heads ~ 25 cm
  - Ears ~ 4 cm
  - Not much multiple scattering
- Scattering off surroundings:
  - Rooms ~ 2m – 10m
  - More multiple scattering
  - Larger sizes lower frequencies



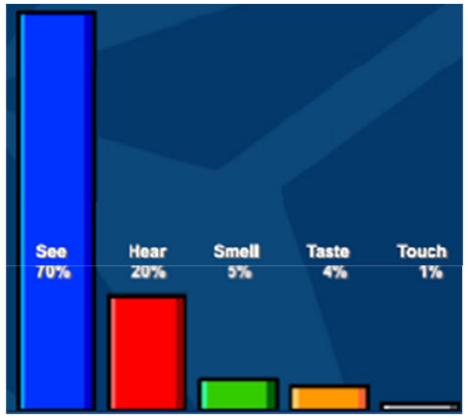
Pinna (Outer Ear) Shape and Hearing

- Pinna (or Auricle):
  - The outer ear with its complex contours
- Low frequency sounds:
  - Pinna behaves similarly to a reflector dish, thus amplifying the sound
- High frequency sounds and Pinna Notch:
  - Some sounds enter ear directly while others reflect off the contours of the pinna
  - Reflections enter the ear canal at a very slight delay
  - This delay results in phase cancellation, where the frequency component whose wave period is twice the delay period is virtually eliminated, called pinna notch
  - This allows us to perceive subtle variations in the locations of sounds
  - Each ear is different, so the effect varies for each individual



Sound in Games

- Role of Sound in Games:
  - "[Sound] serves the story, creates a mood, … and can be the key to bringing the visuals to life" [LoBrutto,1994]
  - 20% of perception is acoustic [Dobbler et. al. 2002]
- Sound Effects:
  - Indicates what is going on around the player
  - May alert player to significant events
  - May include background sounds and musical score
- Narration:
  - Reinforces action
  - Explains non-obvious events and story



Sound Effects

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- Background (or ambient) sound effects:
- Do not explicitly synchronize with the picture— indicate setting
- Examples: Wind, rain, traffic, crowd noises (called “walla”)
- Hard sound effects:
- Common sounds that appear on screen
- Examples: Door slams, weapons firing, and cars driving by
- Foley sound effects:
- Sounds that synchronize precisely with action on screen
- Require the expertise of a Foley artist to record properly
- Examples: Footsteps punching rustling of cloth
- Footsteps, punching, Design sound effects and background score:
- Sounds that do not occur in nature or impossible to record
- Commonly used in science fiction and fantasy genres
- Use of music to create an emotional mood

How can it be done?

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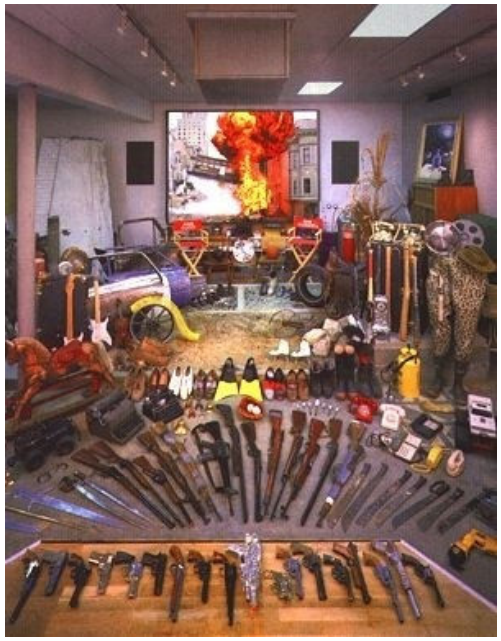
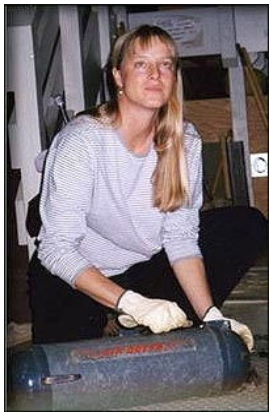
- Foley artists manually make and record the sound from the real-world interaction





Lucasfilm Foley Artist

Foley Artists

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- Ever wondered?
  - Why footsteps sound louder in movies?
  - Why they echo so much?
  - Why the clanging of swords is so ringing and metallic?
- Answer:
  - Foley artists.
  - Named after Jack Foley, a famous 1930's sound engineer.
- Web sites:
  - http://www.marblehead.net/foley/index.html
  - http://www2 digidesign com/digizine/-dz_main.cfm?edition_id=142&navid=1053

Foley Artists

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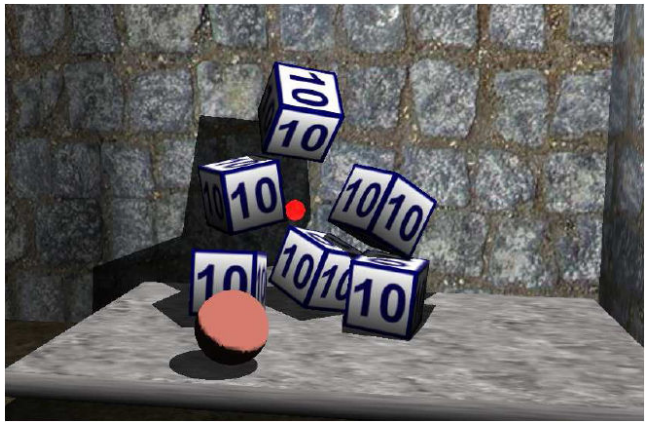
- Effect:
  - Galloping horses
  - Kissing
  - Punching someone
  - High heels
  - Bone-breaking blow
  - Footsteps in snow
  - Star Trek sliding doors
  - Star Wars sliding doors
  - Bird flapping its wings
  - Grass or leaves crunching
- How it might be made:
  - Banging empty coconut shells together
  - Kissing back of hand
  - Thumping watermelons
  - High heels on wooden platform
  - Breaking celery or bamboo or twisting a head of lettuce
  - Squeezing a box of corn starch
  - Pulling a piece of paper from an envelope
  - Flare gun plus sneakers squeak
  - Flapping a pair of gloves
  - Balling up audio tape

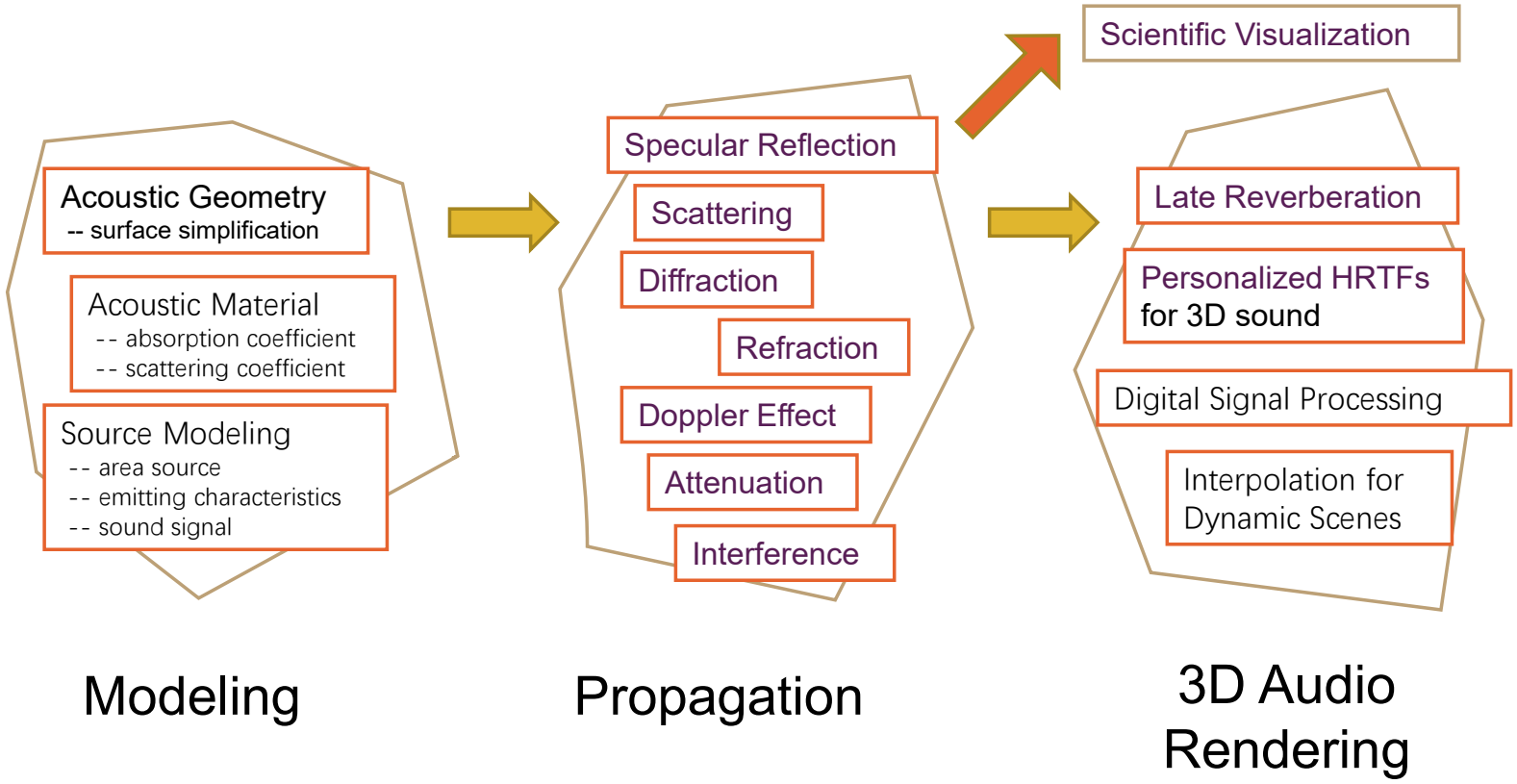
How about Computer Simulation?

- Physical simulation drives visual simulation
  - Sound synthesis can also be automatically generated through physical simulation



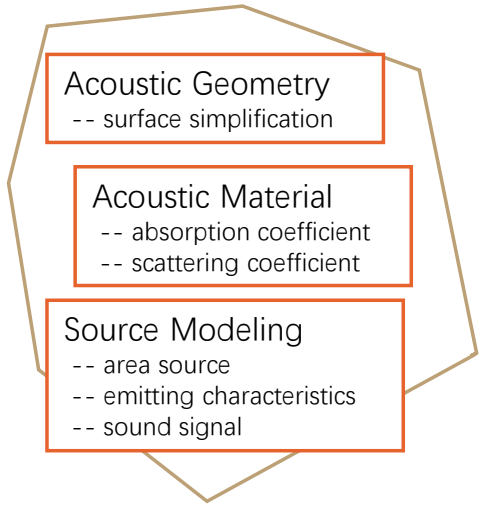
Videos w/ and w/o audio

Sound Rendering: An Overview



Sound Rendering: An Overview

- Modeling
  - Acoustic vs. Graphics
    - Low geometric detail vs. High geometric detail



Modeling

Sound Rendering: An Overview

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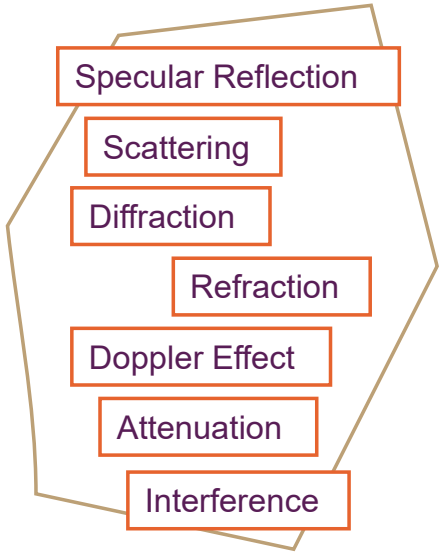
- Propagation
  - Acoustic vs. Graphics
    - 343 m/s vs. 300,000,000 m/s
    - 20 to 20K Hz vs. RGB
    - 17m to 17cm vs. 700 to 400 nm



Propagation

Sound Rendering: An Overview

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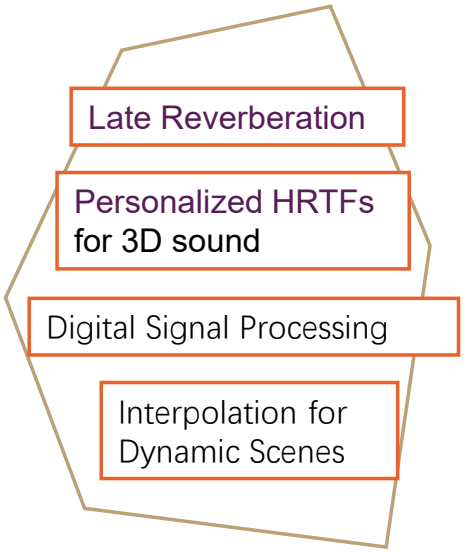
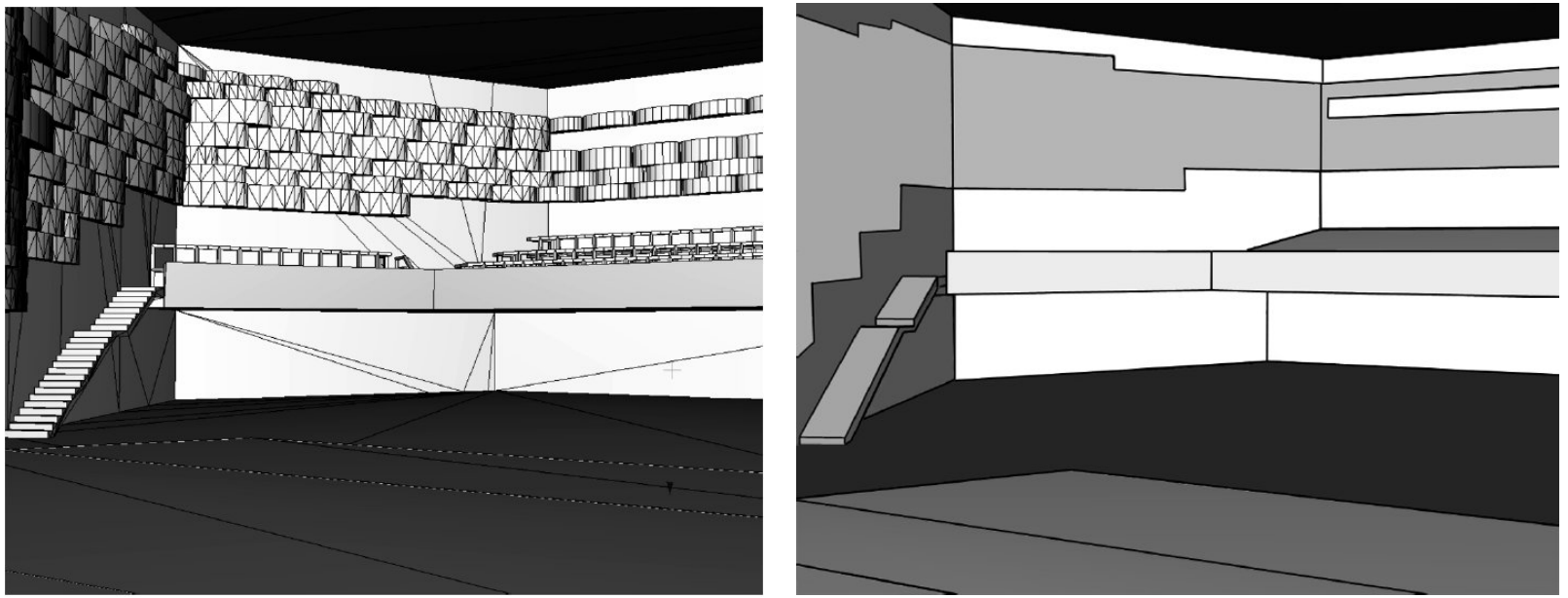
- 3D Audio Rendering
  - Acoustic vs. Graphics
    - Compute intensive DSP vs. addition of colors
    - 44.1 KHz vs. 30 Hz
    - Psychoacoustics vs. Visual psychophysics



3D Audio Rendering

Modeling

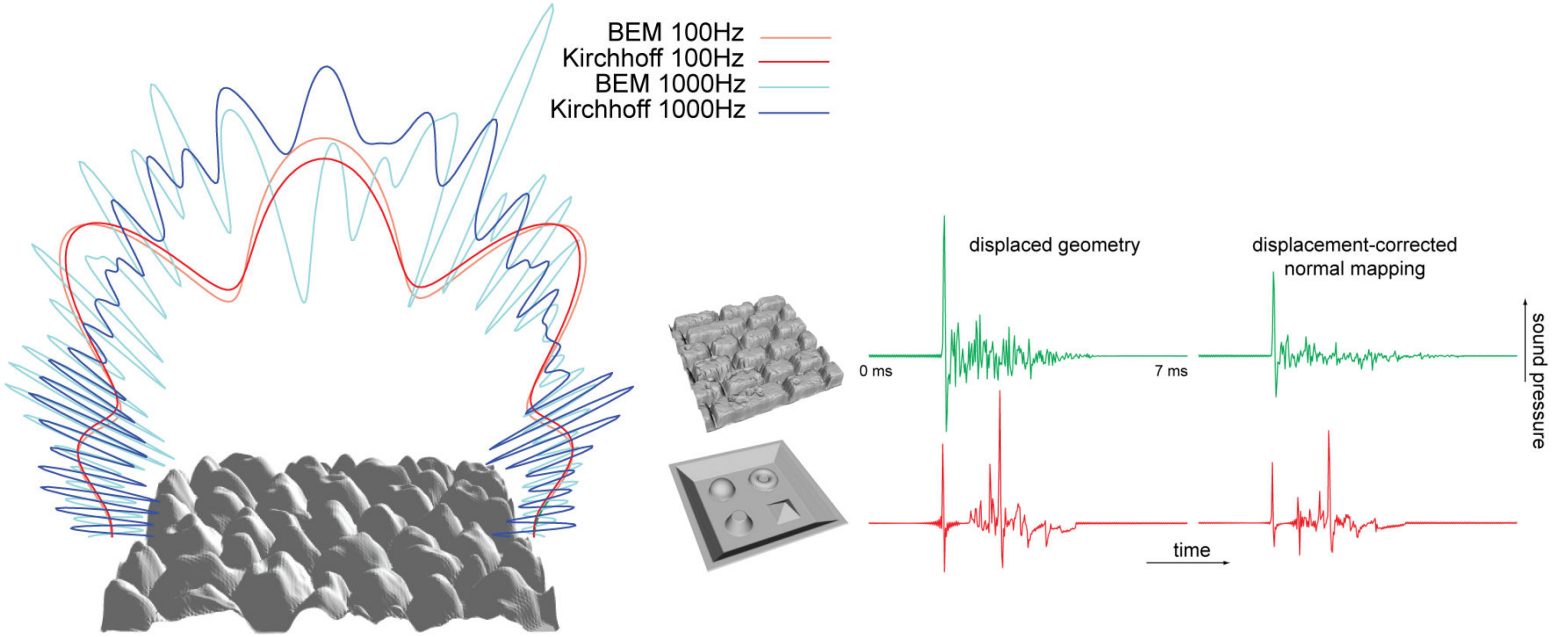
- Modeling Acoustic Geometry



Visual Geometry [Vorländer,2007] Acoustic Geometry

Modeling

- Modeling Sound Material



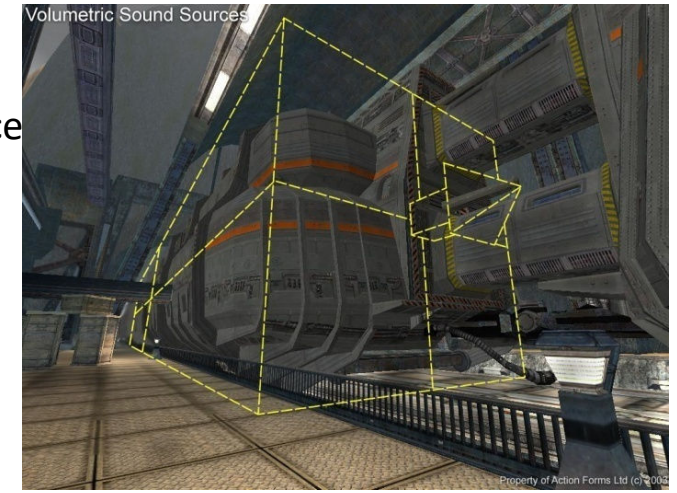
[Embrechts,2001] [Christensen,2005] [Tsingos,2007]

Modeling

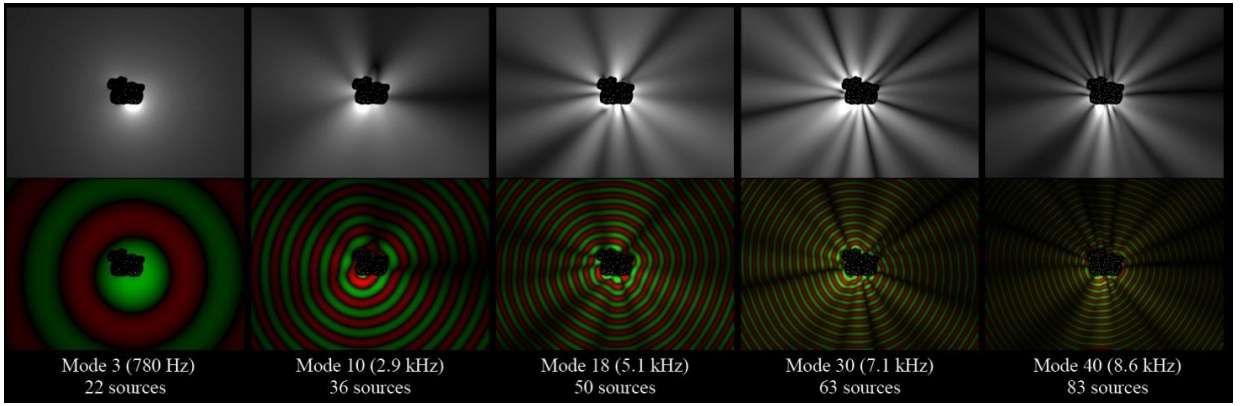
- Modeling Sound Source

Volumetric Sound Source

Directional Sound Source



Complex Vibration Source



Propagation

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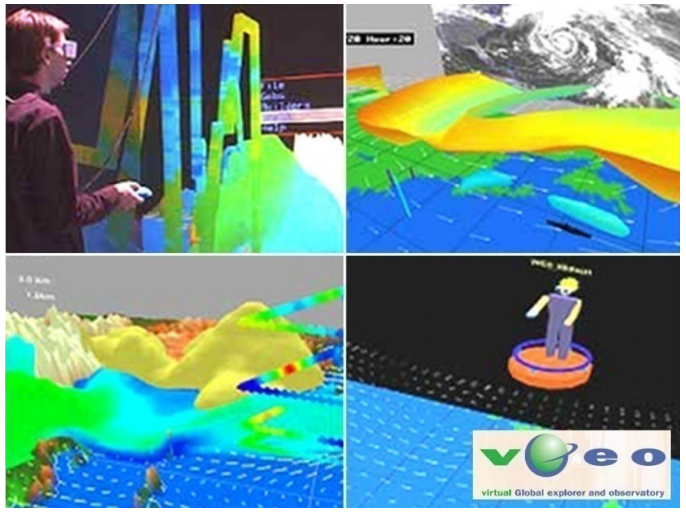
- Applications
  - Advanced Interfaces
  - Multi-sensory Visualization

Minority Report (2002)

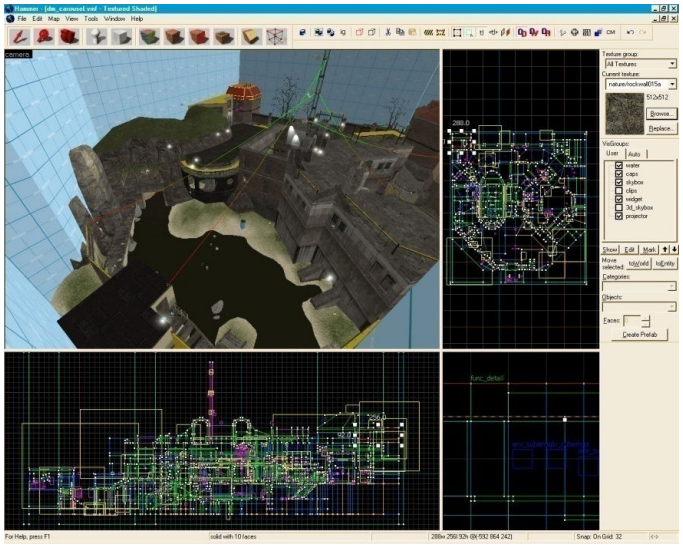


Multi-variate Data Visualization

Propagation

- Applications
  - Acoustic Prototyping

Symphony Hall, Boston



Level Editor, Half Life

Propagation

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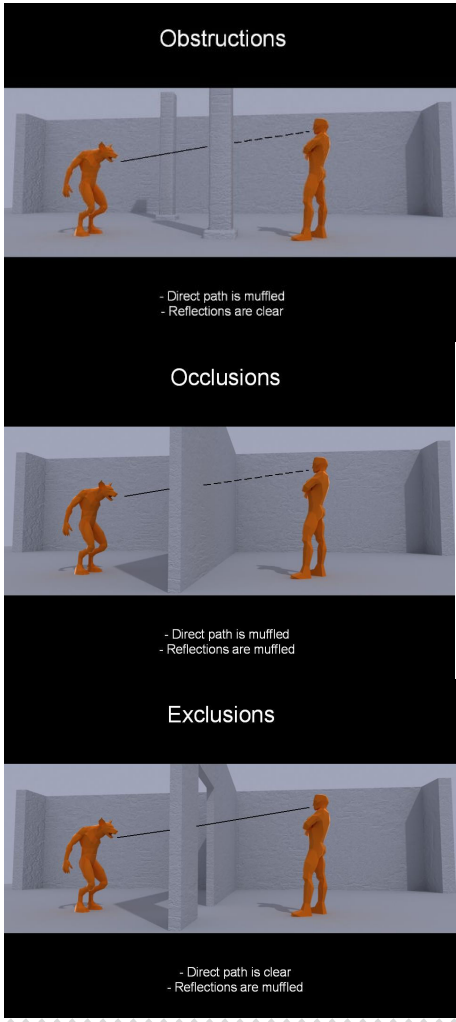
- Applications
  - Games
  - VR Training

Game (Half-Life 2)

Medical Personnel Training

Propagation

- Sound Propagation in Games
  - Strict time budget for audio simulations
  - Games are dynamic
    - Moving sound sources
    - Moving listeners
    - Moving scene geometry
  - Trade-off speed with the accuracy of the simulation
  - Static environment effects (assigned to regions in the scene)



3D Audio Rendering

- Main Components
  - 3D Audio and HRTF
  - Artifact free rendering for dynamic scenes
  - Handling many sound sources

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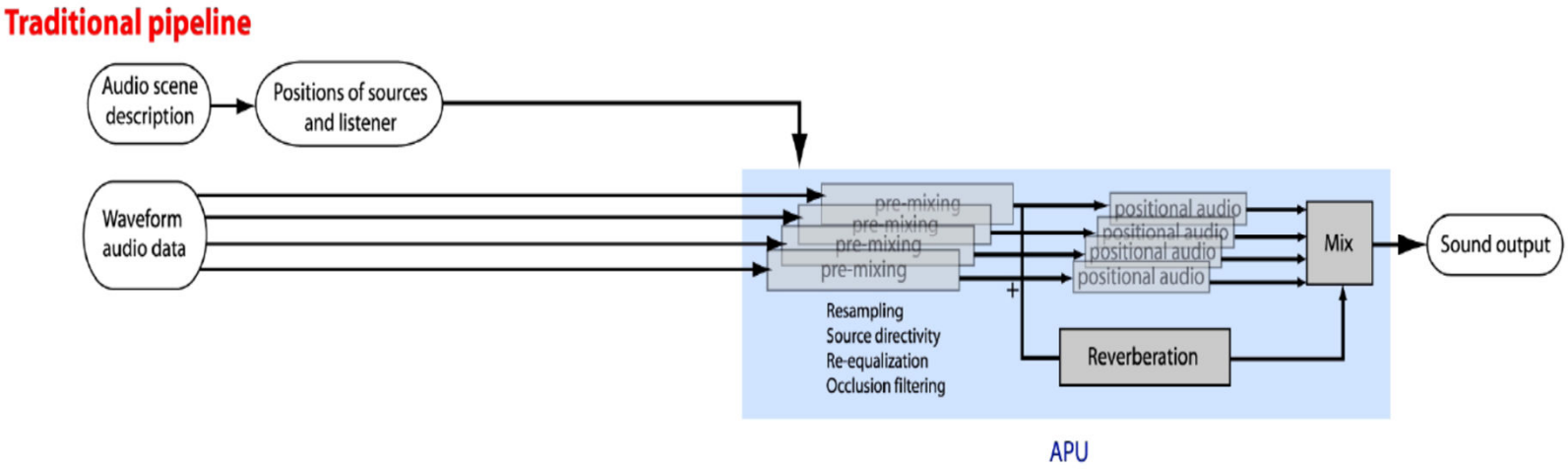

3D Audio Rendering

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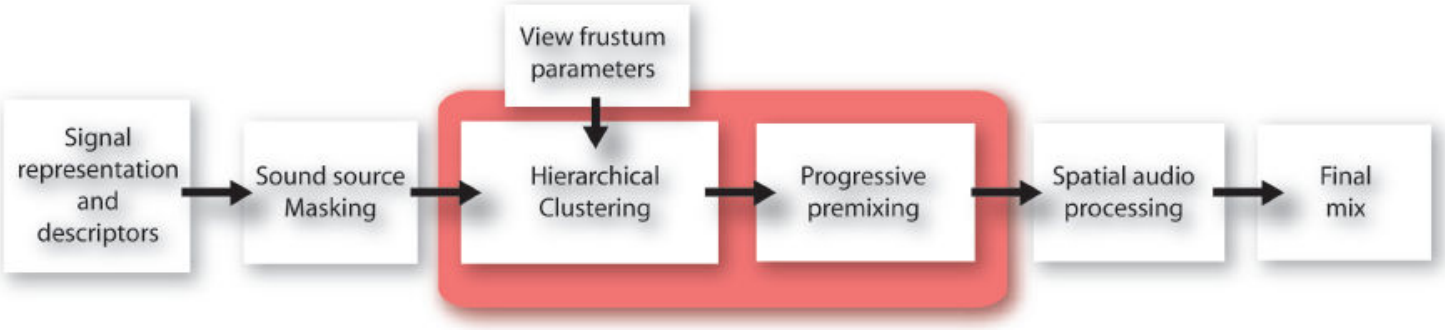
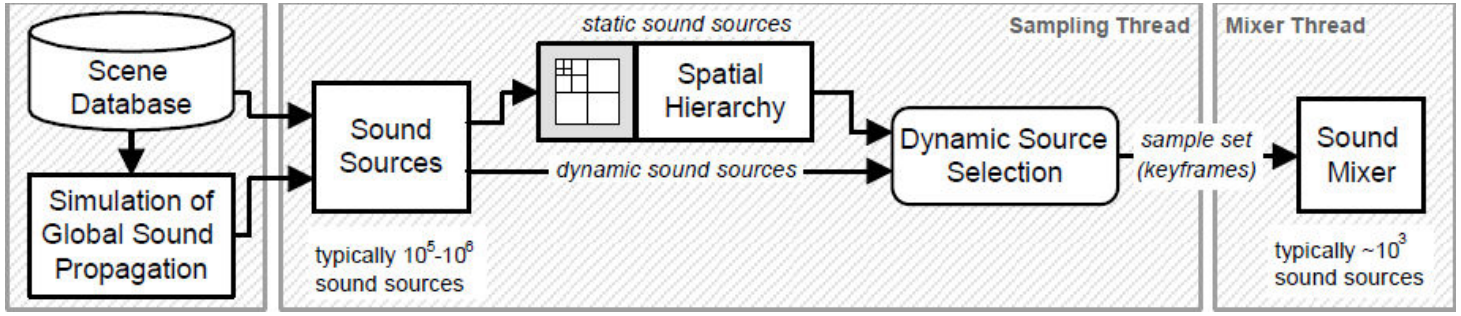
- Perceptual Audio Rendering [Moeck,2007]



- Multi-Resolution Sound Rendering [Wand,2004]



1.2 Sound in VR

--自强不息，止于至善--

VR and sound

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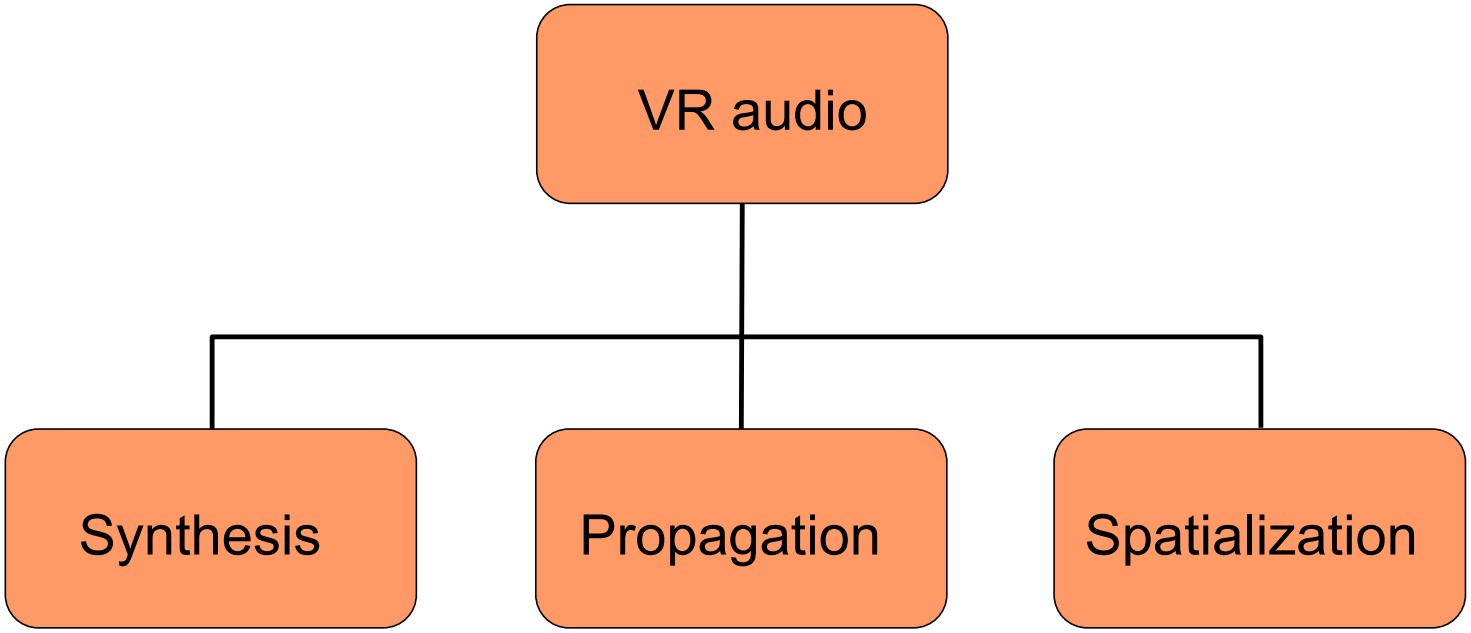
- Sense of presence and immersion
- Situational awareness
- Additional information about environment
- Better audio-visual correlation

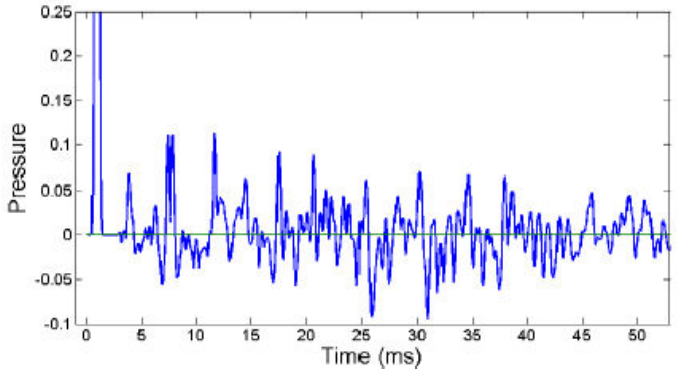
What about VR makes sound different?

- You are part of 3D virtual world rather than watching on 2D screen
- Fidelity and accuracy requirement significantly higher
- Interact with the virtual environment

Physically-based Sound Simulation

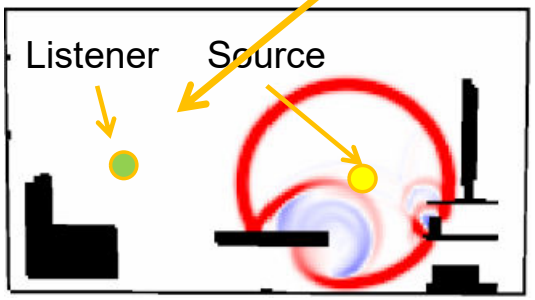


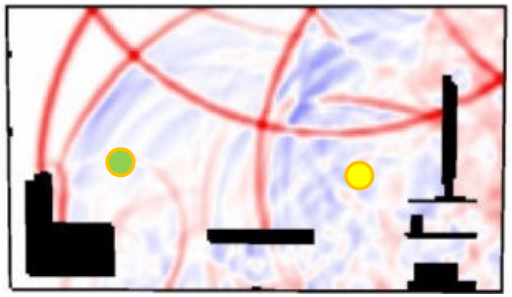
Sound propagation

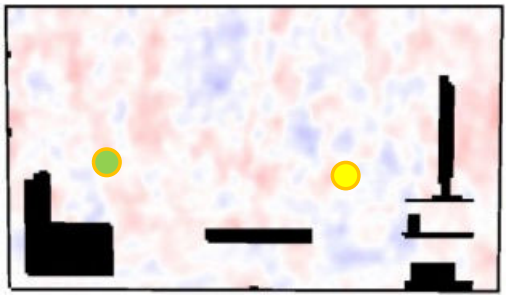


Propagation filter

■High pressure ■Low pressure



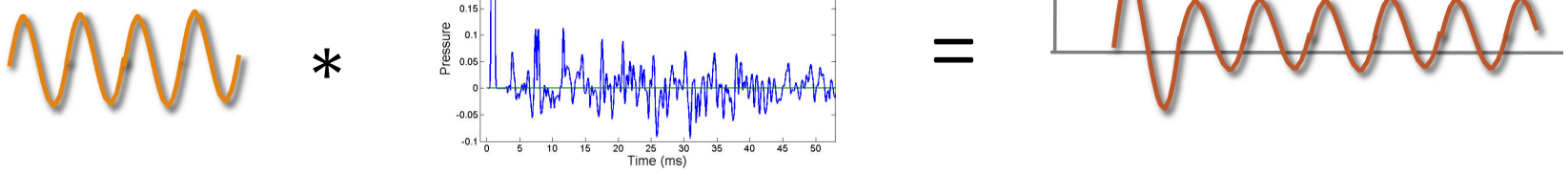


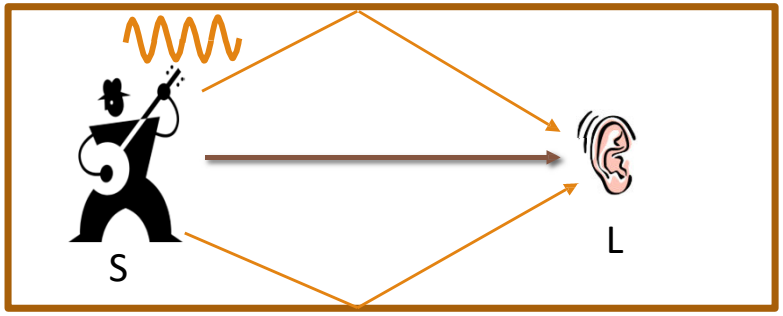


t = 2 ms t = 8 ms t = 50 ms

Sound propagation filter

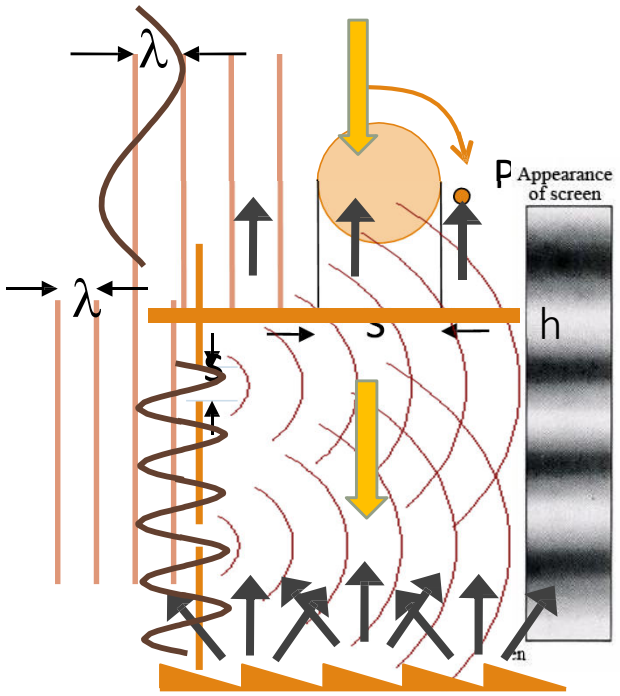
dry audio　　propagation filter　　propagated audio



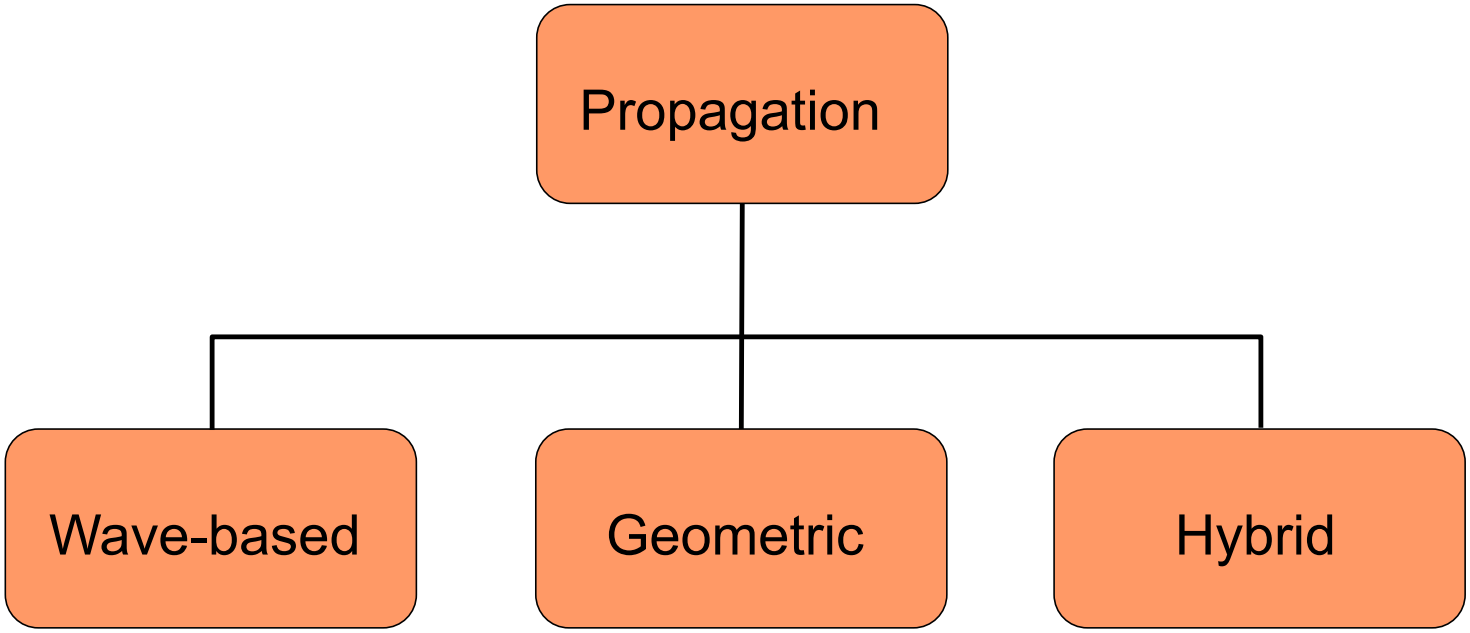


Sound propagation effects

- wavelength ~object size
  - diffraction
  - interference
- wavelength << object size
  - specular reflection
  - scattering



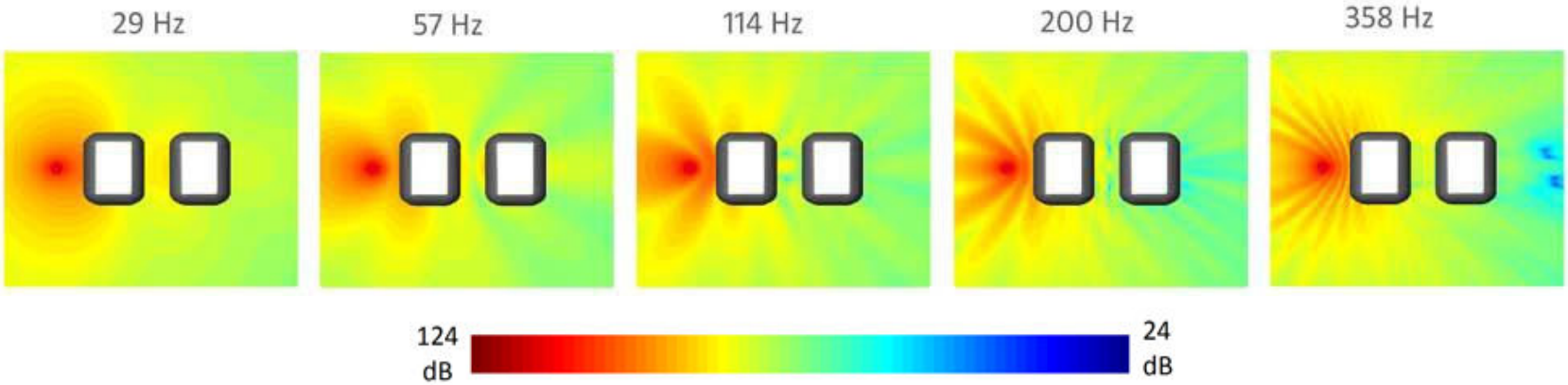
Physically-based Sound Simulation



Wave-based techniques

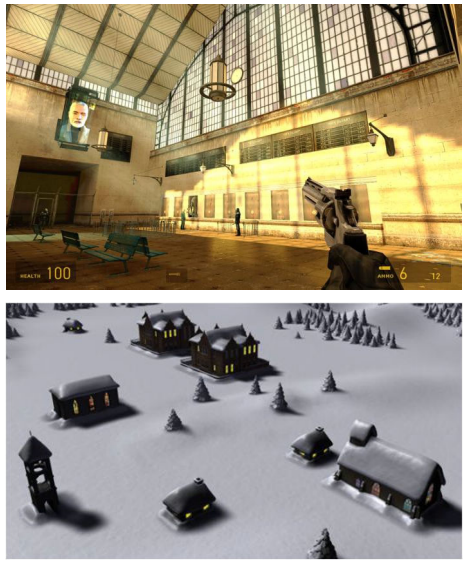
- Solves underlying physics of sound propagation

$$\nabla^2 P + \frac{\omega^2}{c^2} P = 0$$



Wave-based techniques

- Sound radiation
  - precomputed acoustic transfer [James 2006]
- Sound Propagation
  - medium scenes, dynamic sources [Raghuvanshi 2009]
  - large scenes, directivity [Mehra 2014]
  - + dynamic sources [Mehra 2015]



Wave-based techniques

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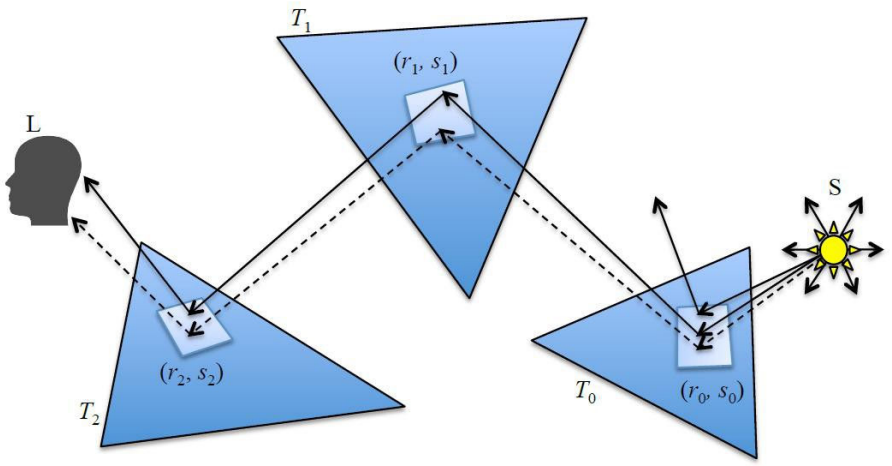
- Accurate at all frequencies
  - reflection, diffraction, interference
- Precomputation $\propto$ freq4
  - hours on CPU cluster (low freq.), days to months (high freq.)
- Interactive runtime
  - 10s of MBs, < 50 ms update rate

Geometric techniques

- Ray-like behavior of sound waves
  - valid for high-frequencies
  - similar to global illumination
  - ray-tracing, visibility graph, etc.



Geometric techniques

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- Typical techniques
  - ray-tracing [Krokstad 1968], image sources [Allen 1979]
  - beam tracing [Funkhouser 1998], phonon tracing[Bertram 2005]
- Approximate diffraction
  - uniform theory of diffraction (UTD)
- Combined techniques
  - ray-tracing, UTD

Geometric techniques

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- Approximate at low, accurate for high frequencies
  - reflection, diffuse reflection
- Negligible precomputation
  - mins on single desktop
- Interactive runtime
  - 10s of MB, < 100 ms update rate

Hybrid techniques

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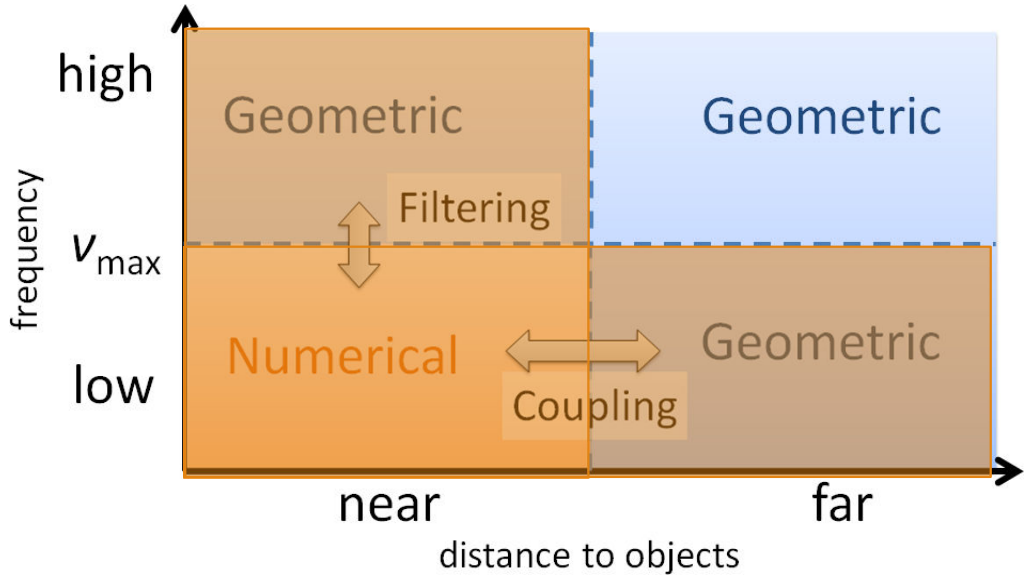
- Wave-based techniques
  - efficient at low freqs., expensive at high freqs.
- Geometric
  - inaccurate at low freqs., accurate at high freqs.
- Hybrid = wave + geometric
  - accurate, efficient for all freqs.

Hybrid techniques

[Hyeh 2013]



Challenges

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- VR game audio
  - source, listener position, scene geometry
  - compute propagation filter dynamically
- Access to dynamic scene geometry

Precomputation budget

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- Sound propagation is computationally expensive
  - multiple freqs., wave-effects
- Analogous to light propagation
  - CPU cluster for light maps
  - tens of hours of cluster time

Runtime resources

- Audio processing
  - < 10% of single CPU, < MBs of memory
- Graphics processing
  - dedicated graphics processor
  - GBs of memory
- Runtime budget
  - at least 1 core, tens to hundreds of MBs

Conclusion

- Accurate sound simulation is important for VR
- Physically-based propagation techniques
  - Wave-based, geometric and hybrid
- Unsolved problems
  - dynamic geometry, pre-computation cost, real-time runtime

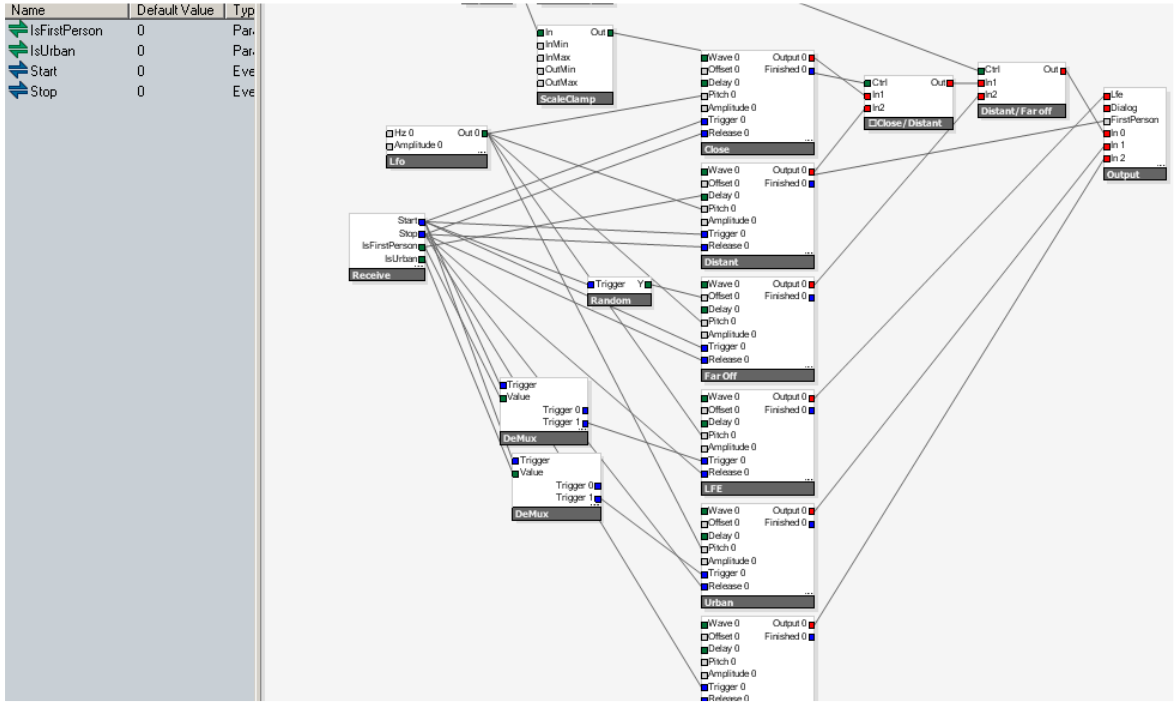
1.3 Integration in to Game Engines

--自强不息，止于至善--

Components of a game audio engine

- Codecs
- Synthesis / Audio « shaders »

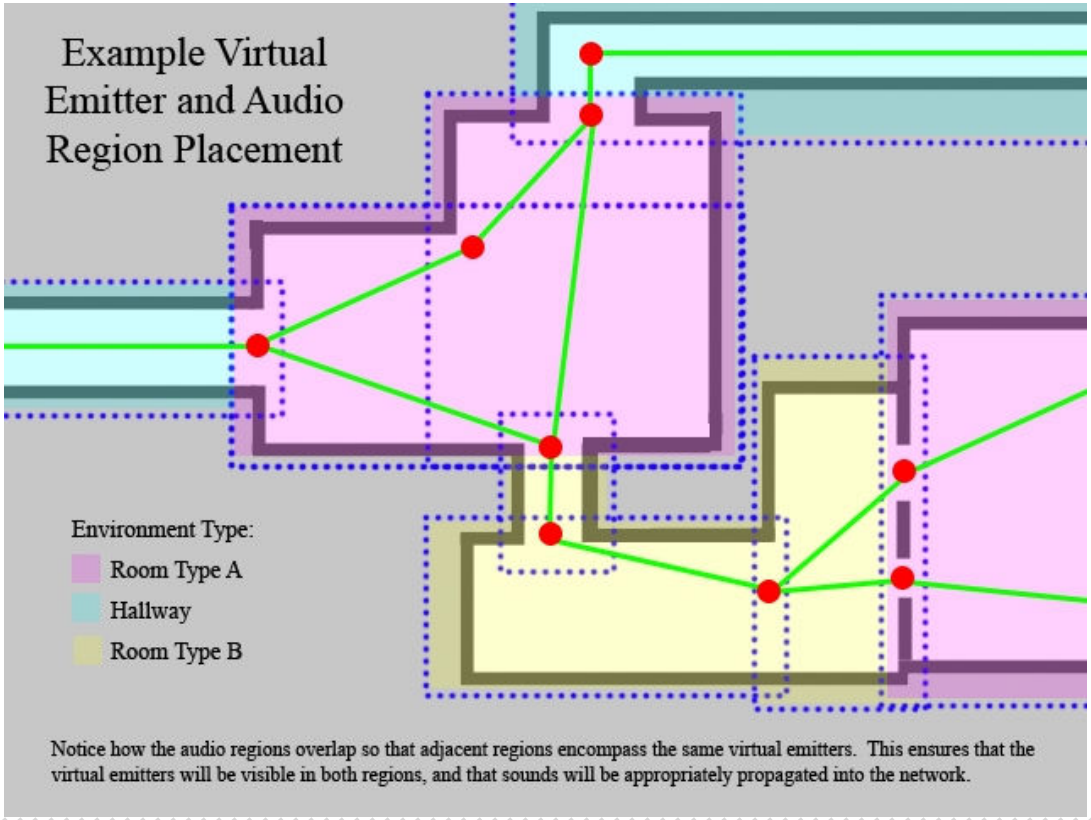
A gun sound “shader” in
Battlefield Bad Company © EA DICE



Components of a game audio engine

- Environmental sound propagation
- DSP effects

Virtual emitter technology for SOCOM confrontation
© SlantSix Games



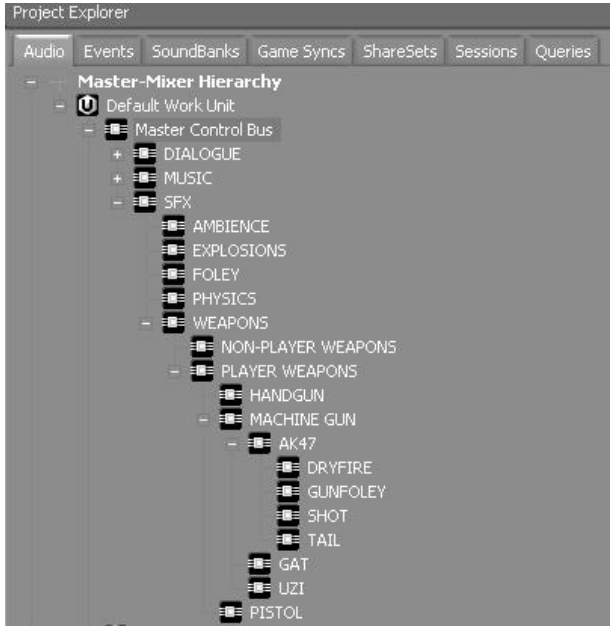
Components of a game audio engine

- 3D audio rendering

Dolby Prologic IIz configuration
with height channels

Components of a game audio engine

- Complex bus routing and mixing
- Mastering, leveling and dynamic range control



Bus grouping in Wwise
© Rob Bridgett

Middleware or in-house ?

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- Powerful middleware available
  - WWise, FMOD, Niles, XAudio, SCREAM replace DirectSound/openAL/EAX
  - real-time processing engines and authoring tools
  - provide visual interfaces to sound designers
  - interactive scripting
  - manage large databases of assets
  - debugging and profiling tools
- State-of-the-art techs still tend to be implemented in-house
  - differentiating factor

Integrating advanced effects in game engines

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- Perceptual rendering
  - clustering and masking

- Frequency-domain pipeline
  - Time vs. frequency domain
  - Geometrical propagation

Clustering for game audio

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- First use of perceptual audio rendering in commercial game engine
  - Next-gen platforms
  - Test Drive Unlimited and Alone in the Dark: NDI

- Test-Drive unlimited data:
  - 200 to 400 sources
  - cars and « physics » clusters
  - error-based refinement
  - 12 clusters sufficient
  - x3 speed-up for 5.1 rendering

Perceptual routing for VoIP

- In-game spatialized chat
  - Optimize bandwidth
  - Use a forwarding bridge

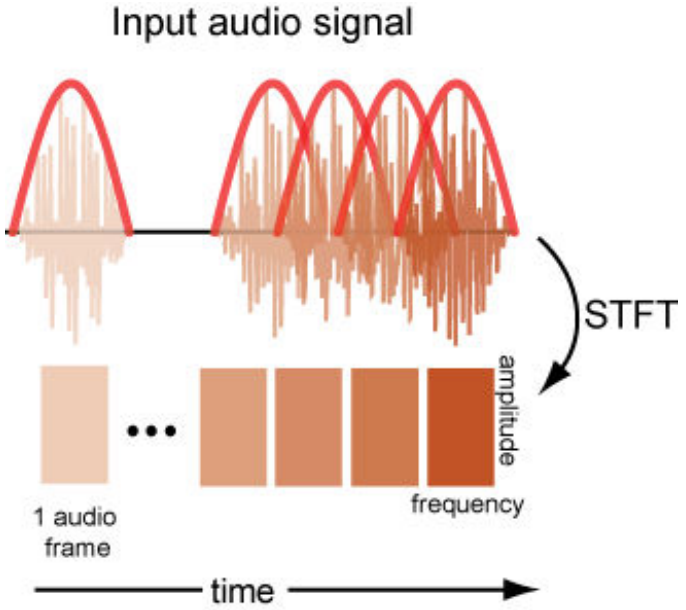
- Bridge performs on the fly masking
  - 70% audio frames discarded
  - Clients perform additional masking+clustering with other in-game sounds

Time vs. frequency domain processing

- Most game audio pipelines perform time-domain processing
  - IIR filtering, mixing, pitch-shifting
  - Fourier-domain FIR convolution
- Advantages of time-domain processing
  - sample-accurate
  - pitch-shifting
  - no transforms
- Advantages of frequency domain
  - arbitrary filtering, subband processing is possible (and fast)
  - progressive/level-of-detail rendering
  - integrates with the codec !

Implementing a frequency domain rendering pipeline

- Choose your transform
  - FFT vs. DCT
  - can be built-in your codec !
- Processing and mixing
  - EQ/convolution/granulation
  - pitch shifting is harder !
  - Zero-padding might not be necessary
- Overlap-add reconstruction
  - apply windows twice before and after the transform (e.g., sine)
  - one inverse transform per output channel is OK



Frequency domain pipeline for geometrical acoustics

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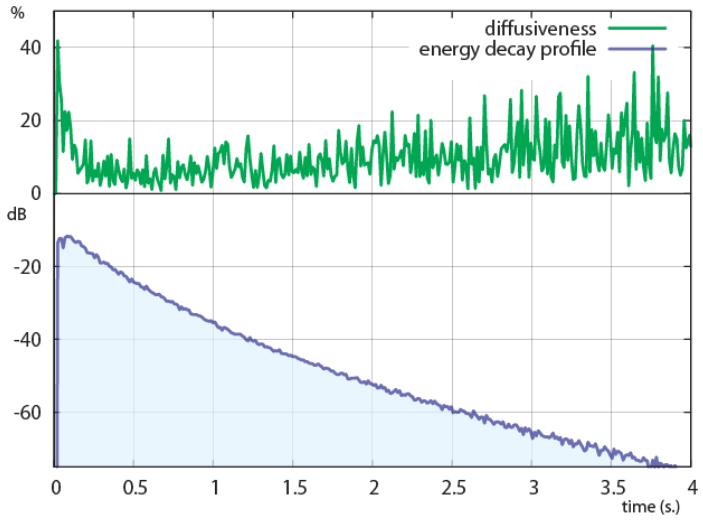
- Reflections and late reverberation
- Early reflections as image-sources
- Late reverberation as parametric model
  - pre-computed
  - hybrid directional/diffuse rendering model



What next ?

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- Meta-data handling
  - computational decision making and performance
  - improve mix quality and creativity
- Integrated decoding, processing and mixing
  - perceptual processing
  - multi-core architectures
- Parametric spatial audio coding [DirAC, SAOC, MPEG surround]
  - Authoring tools ? Recording techniques ?
- Massively multiplayer games
  - audio rendering for thousands of clients (e.g., Dolby Axon)

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2 网 络

--自强不息，止于至善--

2.1 Multiplayer Games and Networking

--自强不息，止于至善--

Multiplayer Games

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- Multiplayer Games: A number of players communicating through a network
  - Persistent Games: Such as “World of Warcraft”, where state is maintained, regardless whether anyone is playing
  - Transient Games: Only exist while people are playing, and reset each time the server-side is reset
- Performance Issues:
  - Latency: How much time delay until global state to be updated?
  - Reliability: How often is data lost or corrupted?
  - Bandwidth: What is the rate of data transfer?
  - Security: How is the game-play protected from tampering/cheating?
- Tradeoffs:
  - All of these considerations interact, and trade-offs must be made

Multiplayer Factors: Event Timing

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- Factors in Multiplayer Games:
  - Event Timing: How do player's interact with the game?
  - Shared Display: How do different players perceive the game state?
  - Connectivity: How are players connected and how do they communicate/interact?
- Event Timing:
  - Turn-based:
    - Player's move in turns (round robin). Other players wait
- Real-time:
  - Players move simultaneously
  - May need to handle race-conditions (e.g., two players attempt to acquire the same resource at the same time)
  - "Twitch Games" typically require very low latencies, less than 150 ms.

Multiplayer Factors: Shared Display

- Shared Display: for multiplayer games on a single platform
  - Full Screen:
    - Complete Player Visibility: Such as board games. Everyone sees everything at all times
    - Player Funneling: Restricts players to a small region of game space, moving with players
    - Turn-Based Screen Control: Active player controls the viewpoint
  - Split Screen: Each player has a separate portion of display through the use of separate viewports
    - Components: System maintains the following information for each player
    - Camera: Viewpoint
    - Cull Data: What portion of the environment is visible
    - Heads-up Display: Game stats displayed on top of scene
    - Map Data: Centered about the current player
    - Audio Effects: Need to be split among players as well
    - Issues: Maintaining consistency between views. Practical only for few players

Multiplayer Factors: Connectivity

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- Direct Link:
  - What: Connected directly typically through short connections
  - Examples: Serial and USB cable, wireless (infrared and Bluetooth)
  - Pro: Fast, reliable. Con: Few players, only small distances
- Circuit-Switched Network:
  - What: Unshared direct connection between endpoints
  - Example: Traditional public telephone system
  - Pro: Reliable, low-latency. Con: Low bandwidth, few players
- Packet-Switched Network:
  - What: Communication broken into small packets that are routed over a shared network
  - Example: Internet
  - Pro: Many players, large areas. Con: Variations in latency/bandwidth

Networking Basics

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- Network:
  - A group of two or more computers connected together
  - Henceforth we consider packet-switched networks
  - Characteristics:
    - Scale: The area spanned by the network
      - LAN: (Local-area network) E.g., connecting one business or school
      - WAN: (Wide-area network) Connecting computers distributed over an entire city, state, country or the world
    - Topology: How the computers are connected together. Examples: Ring, star, bus, tree
    - Protocol: Agreed upon rules for communicating and routing information. Examples: TCP, IP, UDP, FTP, HTTP
    - Architecture: General communication structure. Examples: Peer-to-peer, client server

Networking Basics

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- Protocol:
  - A convention for routing and transferring data over packet switched networks
  - Communication networks may be unreliable and may connect machines having widely varying manufacturers, operating systems, speed, data formats
  - Issues:
    - Packet sizes: Fixed or variable sized packets?
    - Handshaking: Communication exchange to ascertain how data will be transmitted (format, speed, etc.)
    - Acknowledgements: For receipt of data
    - Error checking/correction: Handling errors in data transmission
    - Compression: Reducing data size due to limited bandwidth
    - Encryption: To protect private data

Networking Basics

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- Packet:
  - The logical transmission unit of a protocol
  - Two parts: Header (information) and payload (data)
  - Example: Quake network protocol packet structure:
  - http://www.gamers.org/dEngine/quake/QDP/qnp.html
  - Simple Example:

```
struct Packet {
                                // Header
short packetLength;    // length of the packet (in bytes)
short packetType;       // E.g. data, control, acknowledgement
int checkSum;             //checksum for error checking Payload
char data[256];           // …the data
};
```

Networking Basics

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- Packet
  - Issues
    - Serialization:
      - Pointers and references cannot be reliably transmitted since they refer to local memory. Convert them to names or indices to an array
      - Abstract data types: Are often based on references for inheritance
    - Endianness: Transmitting multi-byte numbers: 0123
      - Little Endian: Low-order bytes first: 3, 2, 1, 0
      - Big Endian: High-order bytes first: 0, 1, 2, 3

Irrelevant Trivia Alert: These terms come from the novel Gulliver's Travels by Jonathan Swift on a civil war between Lilliputian factions that cracked their hard-boiled eggs from the big end or the little end.

    - Intrinsic Types: Use __int32 rather than int to
    - force compiler to use 32-bit integers
    - Unicode: Better than ASCII for string data

Networking Basics

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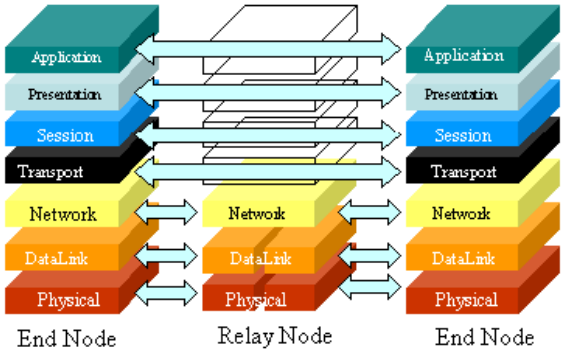
- Open System Interconnect (OSI) Model: Formalizes the multilayered structure of networks
  - Application: End-user processes (e.g., mail (smtp), ftp, telnet)
  - Presentation: Packetization, byte order, encryption, compression
  - Session: Connection and data exchange (e.g., logging in/out, socket)
  - Transport: Flow control (e.g., TCP/UDP)
  - Network: Basic routing (e.g., IP)
  - Data Link: Packet/frame structure
  - Physical: Physical medium wire



Networking Basics

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- Physical Layer
  - The medium over which data is carried
  - Examples: twisted-pair wire, coaxial cable, wireless
  - Latency and Bandwidth:
    - Time of flight: Time to send a single bit of data. Includes delays at switches from source to destination
    - Bandwidth: Maximum transfer rate from source to dest in bits per second (bps). Includes header
    - Transmission time: messageSize / bandwidth
    - Transport latency: timeOfFlight + transTime
    - Sender (receiver) overhead: Time to process packet for sending (receiving)
    - Total Latency: overheadsend + timeOfFlight + transTime + overheadrec
    - Effective Bandwidth: messageSize / totalLatency

Networking Basics

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- Physical Layer
  - Example of Computing Effective Bandwidth:
    - Raw bandwidth: 10Mbps (mega-bits per second)
    - Sender overhead: 250 µsec (0.00025 sec)
    - Receiver overhead: 300 µsec (0.00030 sec)
    - Message size: 1000 bytes (8000 bits)
    - Distance: 1000km (Transmission speed = 150,000km/s)
    - What is the effective bandwidth?

$$TransTime = \frac{8000b}{10Mb/s} = \frac{8000b}{10b/\mu s} = 800\mu s$$

$$TimeOfFlight = \frac{1000km}{150{,}000km/s} = 6667\mu s$$

$$TotalLatency = 250 + 6667 + 800 + 300 = 8017\mu s \approx 0.008s$$

$$EffBandwidth = \frac{8000b}{0.008s} = 1Mb/s$$

    - Note that this is only 1/10th of the raw bandwidth

Networking Basics

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- Physical Layer
  - Maximum bandwidths of common media connection types:

Media Connection Type	Maximum Bandwidth (bps)
Serial cable	20K
USB 1&2	12M, 480M
ISDN	128K
DSL	1.5M down, 896K up
Cable	3M down, 256K up
LAN 10/100/1G BaseT	10M, 100M, 1G
Wireless 802.11 a/b/g	54M, 11M, 54M
Power line	14M
T1	1.5M

Note: Rates vary with direction

Typical Internet Latencies: 50-100ms

Source: Chapt 5.6 of Rabin

  - Actual delivery is around 70% of maximum

Networking Basics

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- Data Link Layer:
  - Puts data in frames and ensures error-free transmission
  - Controls the timing of the network transmission Adds frame type, address, and error control information
  - Examples of data link protocols are Ethernet for local area networks and PPP, HDLC and ADCCP for point-to-point connections
  - Network Interface Card (NIC): Performs these operations. Each NIC is associated with a MAC (media access control) address
  - All devices within a given subnetwork must have unique MAC addresses

Networking Basics

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- Network Layer:
  - Performs end-to-end (source to dest) packet delivery, (whereas the data link layer is for node-to-node)
  - Performs network routing, flow control, data segmentation/de-segmentation, and error control
  - Famous example: The internet protocol (IP)
- IP Addresses:
  - IP version 4: (IPv4)
    - Address is 4 bytes, typically displayed in decimal: 255.8.128.16
    - Only 232 = 4.3 billion possible addresses
  - IP version 4: (IPv6)
    - Address is 16 bytes, displayed as 8 16-bit segments in hex: [2001:0db8:85a3:08d3:1319:8a2e:0370:7344]
    - Supports 2128 = 3.4 x 1038 addresses. Roughly 1 address for every atom in everyone's body on the planet

2.2 Networking and MMOGs

--自强不息，止于至善--

Games and Networking

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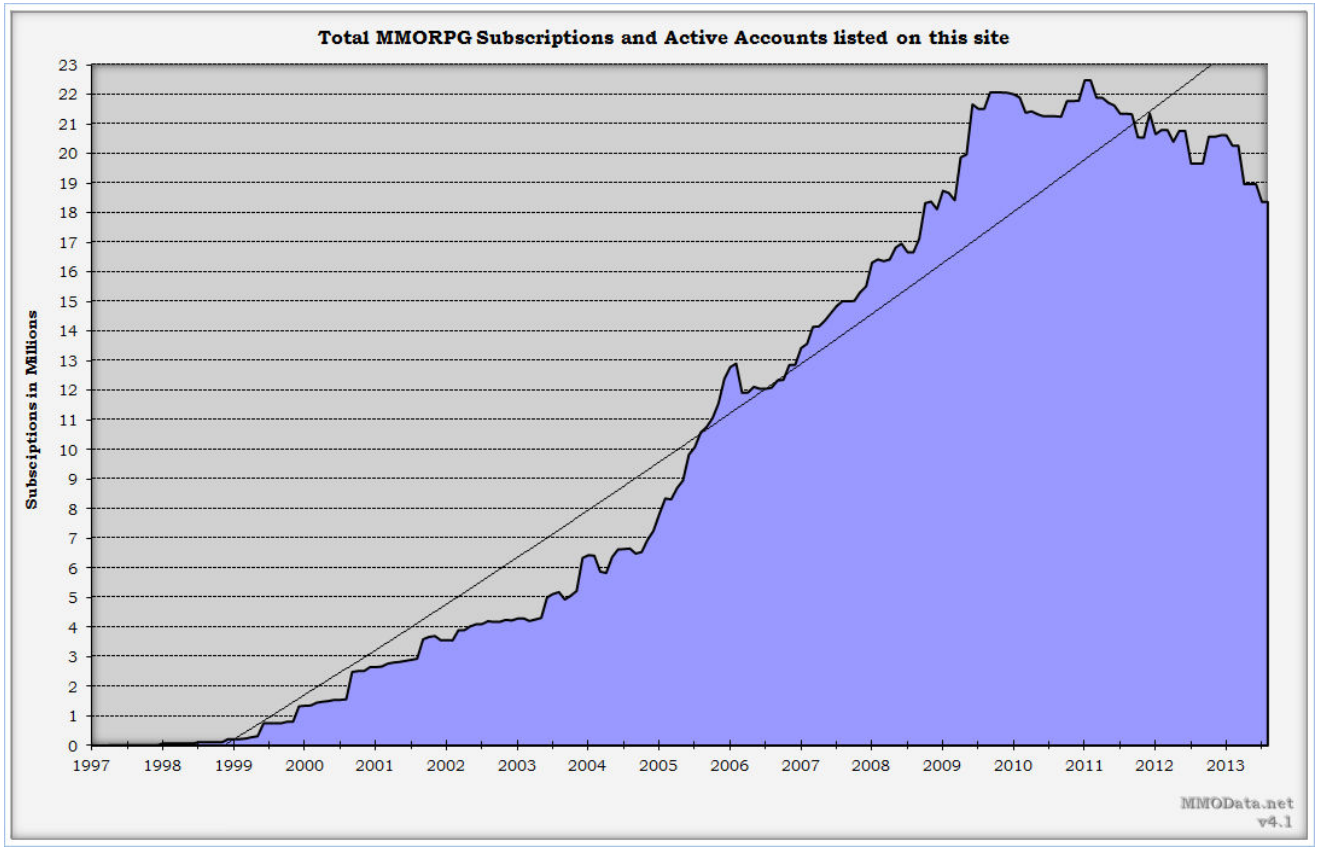
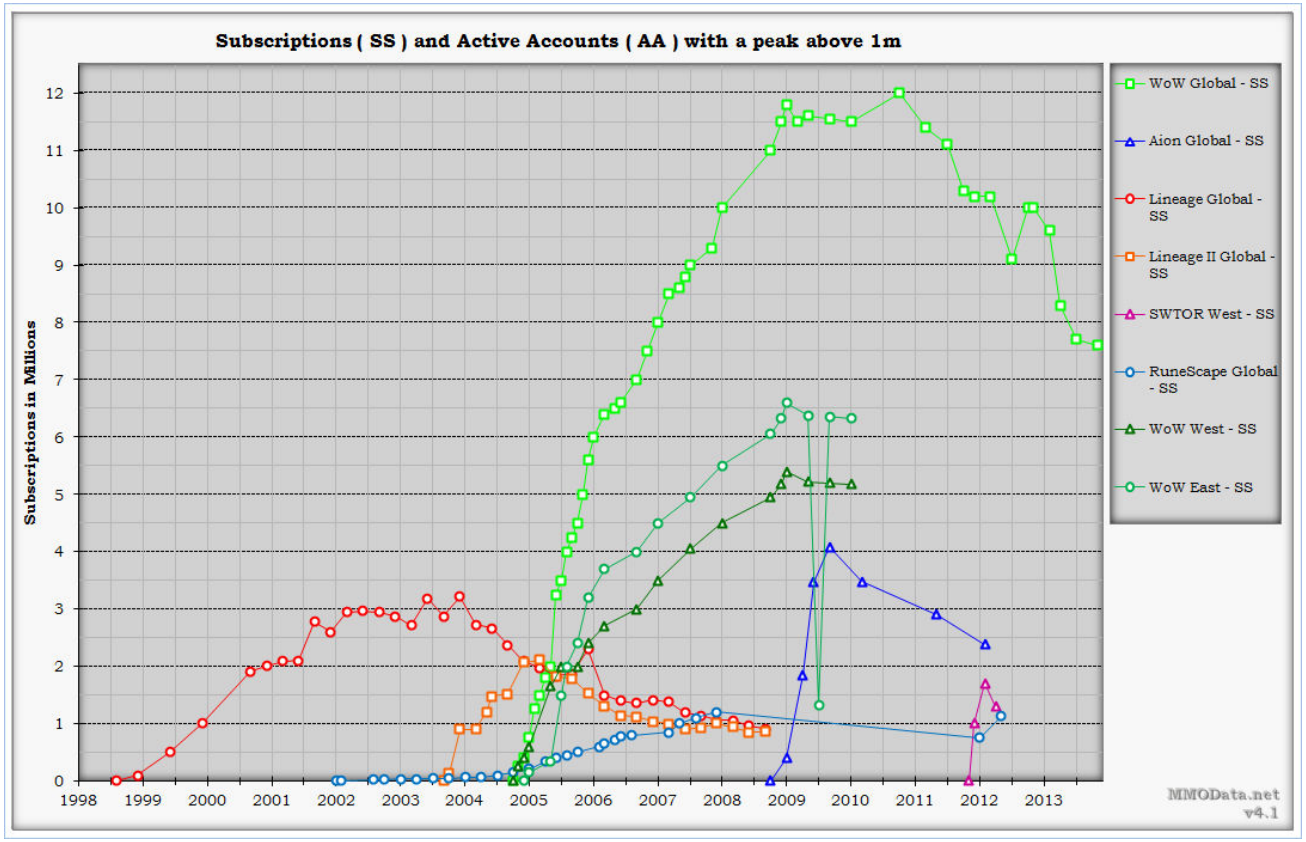
- Industry Information:
  - The top 17 game companies brought in $33B in revenue in 2005
  - Major Players: Combined share $23B
    - Nintendo
    - EA
    - Sony (games part)
    - Microsoft (games part)
- Online Games: About 50% of this is from online games:
  - Massively Multiplayer Online Games (MMOGs)
  - Smaller online play systems: Squad-on-squad play (under 32 players), or player-on-player games
  - Massively Multiplayer Online Role-Playing Games (MMORPGs) about $4.3B

MMOG Subscription Growth



MMOG Subscription Growth



MMOG Rationale

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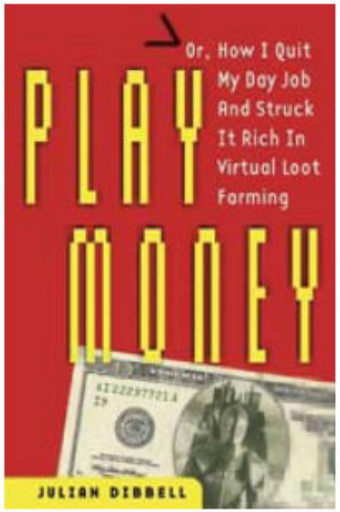
- Why are MMOGs Popular?
- Humans are better at strategy than AI engines are:
  - Less predictable for general settings
- Social Interaction:
  - Peer group support
  - Natural language communication
  - Development of game “culture”
- Make money:
  - Professional gamers
  - Virtual economies
  - …and the inevitable rise in “virtual crime”



Origins of MMOGs

- Quake (1996) :
  - First widely used 3D multiplayer online game
  - Difficult to find game servers:
    - Gamers exchanged IP addresses by email or gaming websites
  - No persistent state:
    - Short-lived ad-hoc fight-to-death sessions
- Advent of MMORPGs:
  - The Realm Online, Meridian 59, Ultima Online, Underlight and EverQuest in the late 1990s
- Scale:
  - 1995: 500 simultaneous players
  - 2000: Thousands
  - 2007: Hundreds of thousands

Persistent MMOGs

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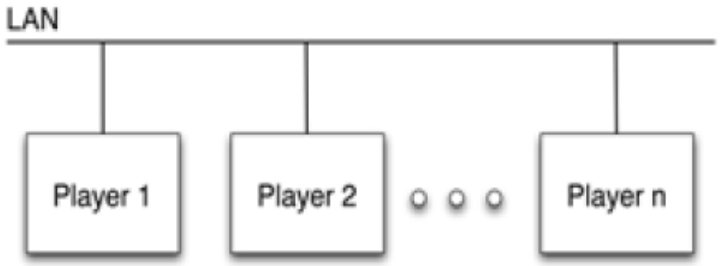
- Need to support millions of subscribers:
  - …a few 100K concurrently
- Back-End Networking:
  - Authentication and billing
  - Ranking / black lists
  - Run-time mods/patches
  - Guard against denial of service attacks
- Front-End (in-game) Networking:
  - Network topology (client-server, peer-peer)
  - Persistence
  - Latency, bandwidth
  - Virtual economy (audits, gold farming, …)
  - Distributed protocols

Online Game Architectures

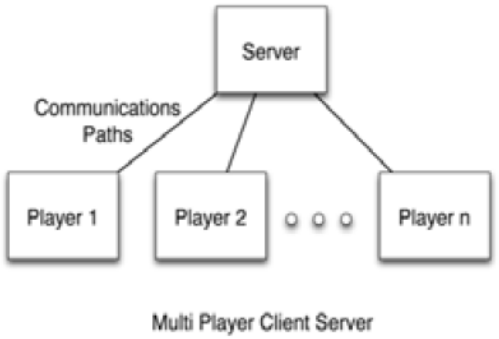
- Peer-to-peer Architecture:
  - Each player communicates directly with all other players
- Possible complexity:
  - O(n2)
  - Limited scalability
- Advantages:
  - Low latency, robust
- Disadvantages:
  - Demands high bandwidth, limited scalability
  - Persistence vs. redundancy tradeoff



Multi-Player Peer to Peer

Online Game Architectures

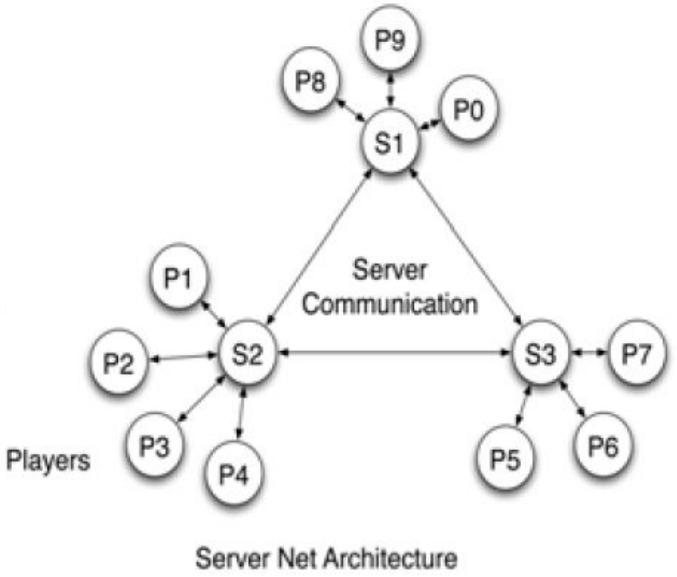
- Client-Server Architecture:
  - Each player (client) communicates with server
- Advantages:
  - Scalable, usually requires less bandwidth.
  - Easier to provide persistence
- Disadvantages:
  - Server bottleneck: Higher latency
  - Server failure: Lower reliability
- Client Types:
  - Thin Client:
    - All simulation on server
    - Good for resource-poor clients: Cell phones, PDAs
  - Simulation Client:
    - Most simulation and world distributed amongst clients
    - Server maintains/updates state



Multi Player Client Server

Online Game Architectures

- Multiple Server Architecture:
  - Many distributed servers, each supporting a subset of the clients
- Advantages:
  - Reduced latency
  - Scalability to millions of players
  - Improved robustness
- Disadvantages:
  - Difficult to maintain consistency for persistence
- Possible Architectures:
  - Shards
  - Mirrors
  - Grids



Server Net Architecture

Online Game Architectures

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- Multi-Server Shards:
  - Each server simulates a different instantiation of the world
  - Each player plays within his own realm, and does not influence the other worlds
  - Minimal server-server traffic.
- Origin:
  - "…different images of the world, trapped in the shattered shards of a mystic gem…" Ultima Online story
- Examples:
  - Ultima Online
  - Neverwinter Nights
  - Silkroad Online
  - World of Warcraft (called Realms)

Online Game Architectures

- Multi-Server Mirrors:
  - Each server mirrors/replicates the same persistent world
- Advantages:
  - Common world view regardless of hosted server
- Disadvantages:
  - Heavy server-server traffic required for synchronization
- Examples:
  - Mankind
  - PlanetSide
  - A Tale in the Desert
  - Entropia Universe

Online Game Architectures

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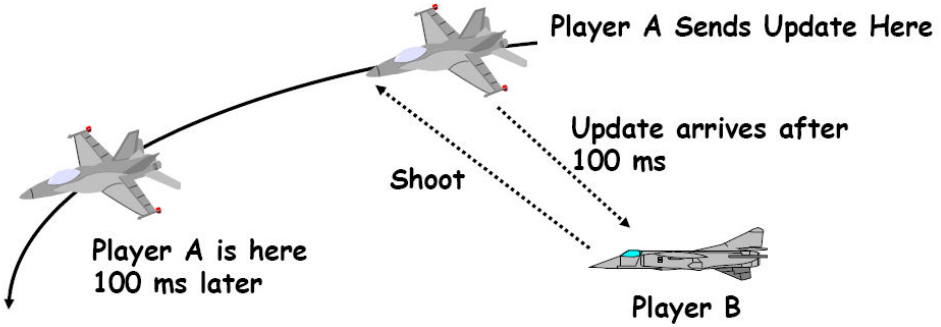
- Multi-Server Grid:
  - Each server (Sim) hosts a different region of the same persistent world
  - Sims communicate with each other on a grid structure
  - Users are transferred between servers as they move around
- Advantages:
  - Scalability for very large worlds
- Disadvantages:
  - Load balancing for high density regions
- Examples:
  - Second Life

Distributed Virtual Worlds

- Implementation is distributed, but appearance should be local
- Need to hide artifacts that suggest otherwise
- Each player sees real-time activity and reacts
- Actions should have immediate impact on the virtual world
- Major Challenge:
  - Latency: Delays induced by network structure



Distributed Virtual Worlds

- Latency:
  - All information received is out-of-date
  - Delays are variable and unpredictable
  - Maintaining the illusion of a consistent and persistent world is much harder when players interact
- Collisions:
  - Simple: Bullets, missiles (linear or parabolic)
  - Complex: Sword fighting, handshakes
  - Nightmare: Successive/concurrent collisions/responses

Distributed Virtual Worlds

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- Visual latency:
  - 30Hz or higher (~ 30 ms)
- Response times: Depend on interaction speed
  - Real-Time Strategy: < 250 ms preferred, 250-500 ms playable, > 500 ms noticeable
  - First-Person Shooter: < 150 ms preferred
  - Car race games: < 100 ms preferred; 100 – 200 ms sluggish; > 500 ms out of control
- Predictability:
  - Predictable (even if sluggish) response considered better than variable (even if sometimes very fast)
- Haptic (touch) latency:
  - Preferred: Update rates > 1 KHz (< 1 ms) in force-feedback systems

Distributed Virtual Worlds

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- Collision detection and response:
  - Did the collision occur?
  - If it did, when?
  - Velocities, forces, in play at that instant
- Environmental effects in a changing world:
  - Global illumination
  - Aural (sound) rendering
  - Both can be computationally intensive

Distributed Virtual Worlds

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- Challenges: Managing Shared States
  - Shared repositories:
    - All servers maintain a common consistent description of the world
    - Absolute consistency!
  - Blind Broadcasts:
    - Owner of each state transmits its current state at regular intervals
    - Clients cache the most recent update information
    - No acknowledgements
    - Frequent state updates compensate for lost packets (hopefully)
  - Dead Reckoning:
    - Transmit state updates occasionally
    - Extrapolate from past updates to estimate the true shared state
    - Need to correct state when an update is received

2.3 Socket Programming and Cheating

--自强不息，止于至善--

Sockets Programming

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- Sockets:
  - Provide a means for programs (running perhaps on different machines) to communicate with one another
  - Types:
    - Stream sockets: (TCP: Transmission Control Protocol)
      - Reliable two-way connected communication streams
      - Items arrive in the order they were sent
      - Virtually error-free
    - Datagram sockets: (UDP: User Datagram Protocol)
      - No connection – Just generate a packet and send it
      - Packets may not arrive in the order they were sent
      - Packets may not arrive at all

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User Datagram Protocol (UDP)

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- Most real-time games use the UDP protocol:
  - Faster and with lower overhead than TCP. Great!
- But there are consequences: UDP packets…
  - … are not guaranteed to arrive
  - … are not guaranteed to arrive in order
  - … are guaranteed to arrive with correct data, but have no protection from hackers intercepting and changing the data once it has arrived
  - … do not require a connection to be accepted. (Assists cheating. For example, intercept the packet "Give …blah… invulnerability," generate a copy of the message, and send it to the server anytime)
- …and global consequences:
  - Unlike TCP, UDP does not provide flow control or aggregation, and so it is possible to overrun the recipient's capacity

Socket APIs

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- Low-level Socket APIs:
  - Berkeley Sockets API: for Unix
  - WinSock: for Windows
  - Both provide essentially the same functionality
- Basic capabilities: (for Berkeley sockets, but Winsock similar)
  - socket(…): Create a socket (of either type)
  - gethostbyname (…): Map hostname (e.g., “mysite.com”) to IP address
  - bind(…): Connect a socket to a port on your local machine
  - connect(…): Connect to a remote machine and port
  - listen(…): Wait for input to arrive from a socket
  - accept(…): Accept a request from another host to connect to you
  - send(…)/recv(…): Send and receive data
  - sendTo(…)/recvFrom(…): Send/receive (for datagram sockets)
  - close(…): Terminate communication

RakNet: More Reliable UDP

- C++-based, open-source toolkit for (higher-level) UDP-based socket programming
- Supports client, server, and peer-to-peer communication
- Provides a layer over UDP, which addresses many of UDP's shortcomings
- Enhancements:
  - Can automatically resend lost packets
  - Can automatically order packets that arrived out of order
  - Protects transmitted data, and inform the programmer if that data was externally changed
  - Provides a connection layer that blocks unauthorized transmission
  - Transparently handles network issues such as flow control and aggregation (grouping many small transmissions into one packet)

RakNet: More Reliable UDP

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- RakNet Standard Headers:

```
#include “MessageIdentifiers.h”
#include “RakNetworkFactory.h”
#include “RakPeerInterface.h”
#include “RakNetTypes.h”
```

- RakPeerInterface:
- The main RakNet object

```
RakPeerInterface* peer =
RakNetworkFactory::GetRakPeerInterface( );
```

- You will usually only generate one of these
- Base class for more specific objects: RakPeer, RakClient, and RakServer

RakNet: More Reliable UDP

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- Connection as a Client:
  - Start up the network threads

```
peer->Startup( 1 30 &SocketDescriptor( ) 1 )
```

    - 1: maximum number of connections. For a pure client, we use 1
    - 30: thread sleep time (in msec)
      - 0 msec: good for games that need fast responses, such as FPS.
      - 30 msec: good response times with little CPU usage
    - SocketDescriptor( ): specifies the port/IP-address to listen on. Since we want a client, we don't need to specify anything
    - – 1: Force RakNet to use a particular IP as host

RakNet: More Reliable UDP

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- Connection as a Client: (cont.)
  - Connect to the RakNet server

```
peer->Startup( maxConnectionsAllowed 30,
                        &SocketDescriptor( serverPort, 0 ), 1 );
peer-
>SetMaximumIncomingConnections( maxPlayersPerServer );
```

    - maxConnectionsAllowed: Maximum simultaneous connections
    - 30: thread sleep time (in msec)
    - SocketDescriptor: Which port to listen to
    - maxPlayersPerServer: Maximum incoming connections to allow

RakNet: More Reliable UDP

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- Read a Packet:

```
Packet* packet = peer->Receive( );
```

  - Returns 0 (NULL) if no input
  - Data may come from engine or other RakNet instances
  - Packet Structure:

```
struct Packet {
        SystemIndex systemIndex; // Server only: Sender index
        SystemAddress systemAddr; // Who sent the packet
        unsigned int length; // Data length in bytes (Deprecated)
        unsigned int bitSize; // Data length in bits (Use this)
        unsigned char* data ; // The data (Cast as needed)
        bool deleteData; // Internal use
}};
```

RakNet: More Reliable UDP

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- Send a Packet:

```
const char* message = "Hello World";
peer->Send( (char*) message, strlen(message)+1,
            HIGH_PRIORITY, RELIABLE, 0,
            UNASSIGNED_SYSTEM_ADDRESS, true);
```

  - message: data to be sent
  - strlen(message)+1: length of data. Allow one additional byte for string's null ('\0') terminator
  - HIGH_PRIORITY: Packet priority (also "LOW" and "MEDIUM")
  - Reliability options:
    - UNRELIABLE: may not arrive and order of arrival arbitrary
    - UNRELIABLE_SEQUENCED: in order, but may not arrive
    - RELIABLE: guaranteed arrival, but order is not
    - RELIABLE_ORDERED: guaranteed arrival, and in order
    - RELIABLE_SEQUENCED: out of order packets are deleted
  - Final arguments indicate a broadcast to all RakNet systems

RakNet: More Reliable UDP

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- Shutting Down:

```
peer->Shutdown( 300 );
...
RakNetworkFactory::DestroyRakPeerInterface( peer );
```

  - Shutdown: closes the connection. The connection can be restarted, using Startup( )
    - 300: Indicates how long to wait (in msec) for remaining packets to be sent
  - DestroyRakPeerInterface: Shut RakNet down and free all memory
  - For more information:
    - See Raknet Manual and Tutorials. (Warning: Doxygen documentation appears to be out of date.)

Observations About Cheating

- Pritchard's Rules (Gamasutra article):
  - 1. If you build it, they will come—to hack and cheat
  - 2. Hacking attempts increase as a game becomes more successful
  - 3. Cheaters actively try to control knowledge of their cheats
  - 4. Your game, along with everything on the cheater's computer, is not secure — not memory, not files, not devices and networks
  - 5. Obscurity ≠ security
  - 6. Any communication over an open line is subject to interception, analysis and modification
  - 7. There is no such thing as a harmless cheat
  - 8 Trust in the server is everything in client-server games
  - 9. Honest players would like the game to tip them off to cheaters, hackers hate it

Why Care About Cheats?

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- Achieving financial advantage:
  - Competitive games with prizes are the obvious example (casinos)
  - Virtual Economies:
    - People play the game, build good characters, and then auction them on eBay
    - If they can cheat to obtain better characters, they are achieving unfair financial advantage
- Ruining your game play => ruining your profits:
  - Online gaming is big business
  - Players tend to have a strong sense of fairness
  - If they believe they are being cheated they will stop playing and
- Cheating is principally on online issue:
  - Single player cheaters only affect themselves, so who cares?

感谢您的观看

Thank you for watching